The Frinkahedron

Geometry, Algebra, and Topology

John Henry

A monograph on enclosure, symmetry,
and recoverable observation

November 2026

"What if the universe only *looks like* it is expanding?"

Preface

This monograph establishes the geometric, algebraic, and cochain foundations of the boundary $F = \partial(P_6 \times P_6)$ and of its recoverable observations. The volume fixes what a finite record must retain: ambient coordinates, sheet signs, the Clifford seam, and certifying two-faces.

The frinkahedron F is a polyhedral three-sphere. A specified homeomorphism $\Psi : F \to S^3$ carries its two solid-torus pieces to the standard genus-one decomposition of the round sphere, on which two families of great circles carry both finite permutation descriptions and quaternion descriptions. Each observation map is given with its fibers; each fiber is given with the retained datum, the *receipt*, that inverts it. Period records on an explicit cell complex carry the same structure, with spectral constants for their reconstruction.

Proofs of the model-specific assertions are given in full. Standard results are cited with their hypotheses. The finite realization is carried to the recovery of each of its thirty-six states, with exact error margins.

The ambient state space has intrinsic dimension three and ambient dimension four. An orthogonal observation $P : S^3 \to \overline{B}^3$ suppresses the fourth coordinate w; its inverse requires an explicit section parameter $\sigma \in \{+1, -1\}$. Two states with one image are separated by chord distance $2|w|$. Removing both certifying faces of the torus complex adjoins a harmonic direction with zero local residual. A zero residual in an incomplete observation does not imply invertibility: recovery is a function of retained topological and algebraic data. The fourth coordinate, the sheet sign, the seam carried by the factorwise map to the Clifford torus, and the certified faces are data of one kind: orientation and incidence information that a projection forgets and a record carries explicitly.

The volume is the geometric card of a table of three. The middle card is a mechanized construction; the numerical card, the companion tolerances volume, fixes what may be claimed from a reading. Chapter 8 is the record: it replies to the thirty-six demands of the review of the middle card, carries the worked finite state as its register, and reads that state once in a press check. Appendix A records provenance separately from the mathematical development.

Notation and conventions

All spaces, maps, and norms are specified when introduced. The following notation collects the principal objects used throughout the monograph.

Symbol	Meaning
P_6, E_j	Regular unit-circumradius hexagon and its j-th edge.
K, F	Product $P_6 \times P_6 \subset \mathbb{R}^4$ and its boundary $F = \partial K$.
$S^3, \overline{B}^3$	Unit sphere in $\mathbb{R}^4$ and closed unit ball in $\mathbb{R}^3$.
Ψ	The specified homeomorphism $F \to S^3$.
$z = (z_1, z_2)$	Complex coordinates on $\mathbb{R}^4 \cong \mathbb{C}^2$.
η, α, β	Latitude and phase coordinates: $z = (\cos\eta\, e^{i\alpha}, \sin\eta\, e^{i\beta})$.
P, L_σ	Orthogonal observation and its sheet-labelled inverse.
C_+, C_-	Simultaneous-phase and opposite-phase circle actions.
H_+, H_-	The corresponding Hopf observations into S^2.
$T_{\pi/4}$	Clifford torus, where $\lvert z_1\rvert = \lvert z_2\rvert = 1/\sqrt{2}$.
$\mathcal{D}, D, E$	General admitted state set, finite clock state set, and state encoding.
O, R	Observation and receipt (retained lifting datum), Definition 1.1.
d_0, d_1	Coboundary maps of the measured cell complex.
$\Delta_1, \mathcal{H}^1$	Hodge Laplacian on one-cochains and its kernel.
λ, γ	Positive spectral gap and smallest observable harmonic response.
M	Measurement map used to recover the harmonic component.
θ	Six-position clock step $\pi/3$.
S_+, S_-	Finite steps $(j,k) \mapsto (j+1, k \pm 1)$ modulo six.

Euclidean vector norms and their induced operator norms are used unless another inner product is declared. The round geodesic metric and Euclidean chord distance on S^3 are distinguished. Angles are in radians; phase coordinates are understood modulo 2π. A bar denotes complex conjugation, and a star denotes the adjoint for the stated inner products. Integer subscripts on hexagon vertices and edges are understood modulo six.

A *calibrated* description specifies the coordinates, inner products, scales, and observation maps needed for its comparisons. Deterministic error bounds are used throughout the worked realization. A probability distribution is introduced only for the separate information counts in Chapter 4.

Contents

Preface **i**

Notation and conventions **ii**

1 The problem of recoverable observation **1**
1.1 A whole and its observable descriptions . . . 1
1.2 The basic diagram . . . 1
1.3 Principal results . . . 2
1.4 Geometric, algebraic, and topological descriptions . . . 3
1.5 Relation to the literature and organization . . . 3

2 Enclosure and the geometry of projection **4**
2.1 The product enclosure and its cells . . . 4
2.2 From the polygonal boundary to the three-sphere . . . 6
2.3 Intrinsic distance and ambient interior . . . 7
2.4 Orthogonal observation and its inverse . . . 8
2.5 Stereographic coordinates and apparent scale . . . 9

3 Circle actions, winding, and algebraic representation **11**
3.1 The torus coordinates and two great-circle families . . . 11
3.2 Hopf observations and their fibers . . . 12
3.3 The paired reading and its antipodal ambiguity . . . 12
3.4 Quaternion multiplication and the rotation group . . . 14
3.5 Finite clocks and exact intertwining . . . 15
3.6 The sixfold symmetry and its two sectors . . . 15

4 Observation, cohomology, and retained information **17**
4.1 Enclosure and accessible domains . . . 17
4.2 Observation fibers and the recovery problem . . . 18
4.3 Finite ambiguity and approximate readings . . . 19
4.4 A cellular model of the accessible torus . . . 20
4.5 Harmonic representatives and measured periods . . . 21
4.6 Transport of representation and metric . . . 21

5 Stability of observation and reconstruction **23**
5.1 Residual energy and the harmonic component . . . 23
5.2 Period measurements and uniqueness . . . 24
5.3 The torus calculation in full . . . 25
5.4 Perturbations and calibration margins . . . 26
5.5 Composition of operations and the role of order . . . 26
5.6 Comparison through a common observable . . . 27
5.7 Trace errors as inputs to reconstruction . . . 28

6 A finite realization and its reconstruction 29
6.1 States, encodings, and the action of the clocks . . . 29
6.2 Orthogonal observation and the retained sheet . . . 30
6.3 Paired circle observations and their antipodal fiber . . . 31
6.4 Separation and finite decoding . . . 31
6.5 A worked state and a deterministic error bound . . . 32
6.6 A calibrated period record . . . 33
6.7 The combined reconstruction . . . 34

7 Enclosure and recoverability 35
7.1 The completed correspondence . . . 35
7.2 Exact structure and quantitative response . . . 36
7.3 Extensions of the construction . . . 36
7.4 The shell as a slice . . . 37

8 The record and the press check 39
8.1 The record . . . 39
8.2 The thirty-six replies . . . 40
8.3 The lifts . . . 50
8.4 The press check . . . 53
8.5 The hypothesis the apparatus tests . . . 55

A Sources and the episode correspondence 56
A.1 Origin of the construction . . . 56
A.2 Episode concordance . . . 56
A.3 The review and the typeset record . . . 57
A.4 The earlier observation framework . . . 59
A.5 Mathematical construction and source evidence . . . 59
A.6 Stanzas and results . . . 59

Bibliography 62

Chapter 1

The problem of recoverable observation

1.1 A whole and its observable descriptions

Let $\mathcal{D}$ denote the domain of admitted states, X an ambient representation space, and Y an observation space. A description of $\mathcal{D}$ is an encoding $E : \mathcal{D} \to X$; an observation is a map $O : X \to Y$. Distinct descriptions support distinct operations: a surface encloses, a list is permuted, a motion composes with its inverse. Comparison of descriptions is comparison of the maps between their admitted states and of the fibers of their observations.

For the three-sphere the orthogonal image in $\mathbb{R}^3$ is a closed ball. An interior point of the ball has two preimages; a boundary point has one. Three data are therefore distinct: the topology of the whole, the image of its observation, and the information required to lift that image to a state. The construction below carries all three explicitly. The whole is a polyhedral boundary $F \subset \mathbb{R}^4$, identified with the round sphere; its motions include two circle actions; its finite realization consists of two six-position clocks; and a measured torus complex carries a period record with local closure and global period recovery.

The domain of the whole is fixed once. For the continuous geometry it is F, equivalently S^3. For the finite realization it is the thirty-six states of the admitted clock sector, and a statement quantified over that sector is complete on its domain. Every proposition in Chapters 1–7 is quantified over a domain so declared, and no subsequent construction amends it. A larger state set requires a new encoding and a new recovery argument.

1.2 The basic diagram

Retained information is a map $R : \mathcal{D} \to \mathcal{A}$ into an auxiliary alphabet. The data of the problem form the diagram

$$\begin{array}{ccccc} \mathcal{D} & \xrightarrow{\quad E \quad} & X & \xrightarrow{\quad O \quad} & Y \\ & \searrow^{(O \circ E,\, R)} & & & \\ & & Y \times \mathcal{A} & \xrightarrow{\quad \mathcal{L} \quad} & \mathcal{D} \end{array} \tag{1.1}$$

in which the reader $\mathcal{L}$ is defined on the image of $(O \circ E, R)$.

Definition 1.1 (Receipt; exact reader)**.** Let $E : \mathcal{D} \to X$ be an encoding and $O : X \to Y$ an observation. A map $R : \mathcal{D} \to \mathcal{A}$ is a *receipt*, or *retained lifting datum*, for (O, E) if $(O \circ E, R) : \mathcal{D} \to Y \times \mathcal{A}$ is injective. An *exact reader* is a left inverse $\mathcal{L} : \mathrm{im}(O \circ E, R) \to \mathcal{D}$,

$$\mathcal{L}\big(O(E(s)), R(s)\big) = s, \qquad s \in \mathcal{D}. \tag{1.2}$$

Proposition 1.2. *An exact reader exists if and only if R is a receipt for (O, E); it is then unique on* $\mathrm{im}(O \circ E, R)$.

Proof. If $(O \circ E, R)(s) = (O \circ E, R)(t)$, applying $\mathcal{L}$ gives $s = t$. Conversely an injective map has a unique inverse on its image. □

A receipt may be a sheet sign, a branch label, a permutation, or an integer retained beside a quotient class; in each case it is an argument of the inverse map. Its size is a property of the observation. A single Hopf observation forgets a circle; the pair of Hopf observations of Chapter 3 forgets an antipodal choice away from two core circles. Extension of $\mathcal{L}$ to data near the image is a stability problem and requires metrics and a reconstruction domain (Chapter 5).

For an observable $q : \mathcal{D} \to V$, recovery of $q(s)$ in place of s requires only that q be constant on every fiber of $(O \circ E, R)$. Agreement of two computed readings is accordingly a statement about their common observable and encoding assumptions; the geometry and the measurement model fix its content [11].

1.3 Principal results

The polytope is

$$K = P_6 \times P_6 \subset \mathbb{R}^2 \times \mathbb{R}^2, \qquad F = \partial K,$$

with P_6 the regular hexagon of circumradius one.

1. *Polyhedral enclosure.* F has f-vector $(36, 72, 48, 12)$; its twelve facets are hexagonal prisms, and the oriented sum of facets is a fundamental cycle (Proposition 2.2).
2. *Seam-preserving sphere structure.* A homeomorphism $\Psi : F \to S^3$ carries $\partial P_6 \times \partial P_6$ onto the Clifford torus $T_{\pi/4}$ and the decomposition $F = A \cup B$ onto the genus-one Heegaard splitting of S^3 (Proposition 2.6).
3. *Observation.* Orthogonal projection has a two-sheet inverse, Lipschitz on every compact subset of the open ball and Hölder of exponent $1/2$ on the closed ball (Proposition 2.10). The paired Hopf observation has fibers $\{z, -z\}$ off the cores.
4. *Winding.* The left and right Hopf actions $C_\pm$ have winding vectors $(1, 1)$ and $(1, -1)$, with $\det B = -2$; they span an index-two sublattice, and the finite steps $S_\pm$ generate an index-two subgroup of $(\mathbb{Z}/6\mathbb{Z})^2$ with two sectors of eighteen states (Propositions 3.3 and 3.4).
5. *Cochain response.* On the two-cell torus complex the Hodge Laplacian is $\Delta_1 = 6P_v$, with spectrum $\{0, 0, 6\}$, spectral gap $\lambda = 6$, and harmonic measurement response $\gamma = 1/\sqrt{3}$ (Proposition 4.6); the complete observation has least singular value $\tau = \sqrt{(7 - \sqrt{41})/2}$ (Section 5.3).
6. *Finite realization.* The thirty-six clock states have minimum image separations $s_P = 1/(2\sqrt{2})$, $s_H = \sqrt{2}$, and $s_E = 1/\sqrt{2}$ (Lemma 6.2); the worked state $(1, 2)$ is recovered exactly under declared bounded errors (Theorem 6.3); five descriptions of the sector are isomorphic under the clock action (Theorem 7.1).

The component theories (spheres, Hopf fibrations, Clifford actions, finite cochain complexes, singular-value estimates) are standard. The contribution is their explicit assembly on one admitted domain.

1.4 Geometric, algebraic, and topological descriptions

Geometry supplies the state space, its metric, and its observation maps. One topological sphere carries many metrics and many maps into $\mathbb{R}^3$. Orthogonal projection and the stereographic chart are accordingly distinct: the former has a compact image and forgets a sheet; the latter omits one point. A drawing is the image of a map, and the map is defined independently of the drawing.

Algebra supplies the operations. Cyclic permutations of a sampled orbit and the quaternion motion through its samples satisfy an exact intertwining identity (Equation (3.4)). An arbitrary permutation need not arise from the chosen motion, and a continuous motion need not preserve a cell decomposition; the subgroup and its action are fixed before an operation is transferred.

Topology supplies the relations invariant under continuous change of coordinates. A loop on a torus may represent a nonzero class while contracting in the containing three-sphere; deleting a core circle changes the admitted contractions. The domain of a period is part of the statement about that period. Cellular incidence makes these relations finite, and positive inner products give them a quantitative Hodge theory [7, 9, 1].

The three descriptions meet in reconstruction. Topology identifies the undetermined harmonic directions; a calibrated algebraic map measures them; geometry and linear analysis supply the constants of error propagation. Exact identities specify the model, and positive lower bounds make its inverse stable.

1.5 Relation to the literature and organization

The sphere, its product decomposition, and the Hopf construction follow [7, 12, 13]. The quaternion and Clifford conventions follow [17, 4]. The cochain discussion uses finite Hodge theory and compatible transport of complexes [9, 1]; singular values supply the inverse estimates [6].

Chapter 2 constructs the polyhedral whole and its projections. Chapter 3 develops the two circle actions, their Hopf observations, and their finite and Clifford descriptions. Chapter 4 relates observation fibers to receipts and locates the periods in an explicit complex. Chapter 5 proves the reconstruction and stability estimates. Chapter 6 treats the finite realization. Chapter 7 states the correspondence theorem and its extensions. Chapter 8 is the record: it replies in order to the thirty-six demands of the review, carries the worked state as its register, and reads that state once in a press check. The appendix records source provenance.

Chapter 2

Enclosure and the geometry of projection

The object of this chapter is a boundary together with its observations: an intrinsic dimension, an embedding, and a family of observation maps. The embedding fixes position, the observation fixes which distinctions survive, and the metric fixes how a perturbation is measured. The polytope supplies twelve facets, their attaching data, and a separating torus; the homeomorphism Ψ carries the boundary onto the round three-sphere and the separating torus onto the Clifford torus.

The four ambient coordinates are configuration coordinates. A physical interpretation requires a calibration identifying the observable they measure; the results of this chapter concern the declared configuration space.

2.1 The product enclosure and its cells

Put $\theta = \pi/3$, and let

$$u_j = (\cos j\theta, \sin j\theta), \qquad P_6 = \operatorname{conv}\{u_0, \dots, u_5\} \subset \mathbb{R}^2, \qquad K = P_6 \times P_6 \subset \mathbb{R}^4.$$

All vertex and edge subscripts are taken modulo six. The regular hexagon contains the origin in its interior. With

$$c = \cos(\pi/6) = \sqrt{3}/2, \qquad n_j = (\cos((j + \tfrac{1}{2})\theta), \sin((j + \tfrac{1}{2})\theta)),$$

its supporting-half-plane description is

$$P_6 = \{u \in \mathbb{R}^2 : n_j \cdot u \le c \text{ for all } j\}.$$

The edge E_j joining u_j to u_{j+1} lies in the supporting line $n_j \cdot u = c$. The twelve inequalities for K separate into six in each coordinate plane; this separation gives both the cell census and the two solid tori of the boundary.

Definition 2.1 (The frinkahedron)**.** The *frinkahedron* is the polyhedral boundary

$$F := \partial(P_6 \times P_6) \subset \mathbb{R}^4$$

of the hexagonal duoprism $K = P_6 \times P_6$. Its *facets* are its three-dimensional cells. A point of F is a complete boundary configuration, comprising both coordinate-plane components.

The vertices of K are the 36 ordered pairs (u_j, u_k). Its facets are

$$A_j = E_j \times P_6, \qquad B_k = P_6 \times E_k.$$

Each facet is a hexagonal prism. Within one family, adjacent edges of the underlying hexagon give adjacent facets: $A_j \cap A_{j+1} = \{u_{j+1}\} \times P_6$, a hexagonal two-cell. Nonadjacent edges give disjoint facets. The same rule holds for the B family. Across the families, every pair meets, and

$$A_j \cap B_k = E_j \times E_k$$

is a square. Intersections of lower dimension are obtained by intersecting the two factors separately; the attaching data are determined by the product.

Proposition 2.2. *The boundary F has f-vector $(36, 72, 48, 12)$. Every two-dimensional face belongs to exactly two facets, and the oriented sum $[F] = \sum_j A_j + \sum_k B_k$ of the facets is a cellular three-cycle. It generates $H_3(F; \mathbb{Z}) \cong \mathbb{Z}$.*

Proof. A face of a product of polytopes is the product of a face in each factor. The proper faces here have dimensions adding to at most three. There are $6 \cdot 6 = 36$ vertices. An edge is an edge in one factor times a vertex in the other, giving $2 \cdot 6 \cdot 6 = 72$. Two-faces are the 36 edge products and the twelve vertex-times-hexagon products. The twelve facets have already been enumerated.

A square $E_j \times E_k$ belongs to A_j and B_k. A hexagon $\{u_j\} \times P_6$ belongs to the two A facets corresponding to edges incident on u_j; the other kind of hexagon has the analogous two B facets. Orient $\mathbb{R}^4$ by its ordered coordinate planes, and give each facet its boundary orientation. The outward normals are $(n_j, 0)$ and $(0, n_k)$. At a shared two-face the two induced boundary orientations are opposite. Each term therefore cancels in the boundary of the sum, so $\partial[F] = 0$. The complex has no four-cells, so $H_3(F; \mathbb{Z}) = \ker \partial_3$. By Proposition 2.6, F is a closed connected orientable three-manifold, so $\ker \partial_3 \cong \mathbb{Z}$; a cycle in which every facet appears with coefficient ± 1 is a generator. □

Dimension	Cell types	Count	Signed count
0	vertex × vertex	36	+36
1	edge × vertex, and reverse	72	−72
2	squares; hexagons in either factor	36 + 12	+48
3	edge × hexagon, and reverse	12	−12

Table 2.1: The cells of F. The Euler characteristic is $36 - 72 + 48 - 12 = 0$. Two-dimensional faces and three-dimensional facets enter different parts of the census.

Remark 2.3. Enclosure is the incidence statement $\partial[F] = 0$: every two-face is shared with opposite orientations. The alternating count $36 - 72 + 48 - 12 = 0$ is a consistency relation and does not identify a space; the sphere topology is Proposition 2.6. No proof in this volume uses a figure, and a named object is the object of its definition.

Lemma 2.4 (Incidence parity)**.** *In a closed polyhedral n-manifold every $(n-1)$-cell lies in exactly two n-cells, so the total number of $(n-1)$-cells counted over the boundaries of all n-cells is even.*

Proof. Each $(n-1)$-cell is counted once for each of its two incident n-cells. □

For $n = 2$, three quadrilaterals and nine triangles contribute $3 \cdot 4 + 9 \cdot 3 = 39$ sides and form no closed surface. For F, $n = 3$: the twelve prisms contribute $6 \cdot (6 + 2) = 96 = 2 \cdot 48$ two-faces, in agreement with Table 2.1.

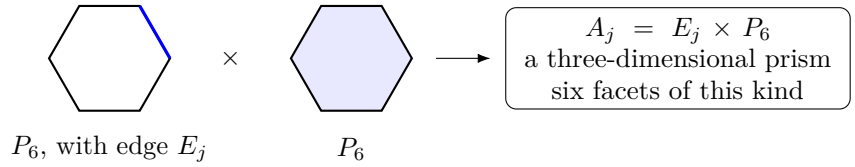


Figure 2.1: A product incidence diagram. Interchanging the factors gives the other six facets. The diagram records a construction of cells rather than a perspective image of the four-polytope.

2.2 From the polygonal boundary to the three-sphere

A convex body containing the origin can be described by its distance to the boundary in each radial direction. For a unit vector $v \in S^1$, that distance for P_6 is

$$r(v) = \frac{c}{\max_j n_j \cdot v}.$$

The six normals are equally spaced. One lies within angle $\pi/6$ of any given direction, and therefore

$$c \le \max_j n_j \cdot v \le 1, \qquad c \le r(v) \le 1.$$

These bounds make the radial change of coordinates continuous at the origin.

Lemma 2.5. *The map $g : P_6 \to D^2$ defined by $g(0) = 0$ and*

$$g(u) = \frac{u}{r(u/\|u\|)} \quad (u \neq 0)$$

is a homeomorphism carrying ∂P_6 onto S^1. It commutes with rotation through θ.

Proof. For a fixed unit direction v, every point of the polygon on that ray has the unique form tv, $0 \le t \le r(v)$. The formula sends it to $(t/r(v))v$. Its inverse on the disk is consequently

$$g^{-1}(sv) = sr(v)v, \qquad 0 \le s \le 1.$$

The radius function is continuous and positive. The displayed uniform bounds prove continuity of both maps at zero; away from zero continuity follows from the formulas. Rotation through θ permutes the normals, leaving their maximum unchanged. Hence r has that rotational symmetry, and so does g. □

Let $G = g \times g : K \to D^2 \times D^2$, identifying each disk with $\mathbb{C}$. It carries the product boundary onto

$$\partial(D^2 \times D^2) = \{(v_1, v_2) : \max(|v_1|, |v_2|) = 1\}.$$

There are two successive normalizations: each polygon is first carried to a disk, and then the product boundary is carried to a Euclidean sphere. Define

$$\Psi : F \longrightarrow S^3, \qquad \Psi(p) = \frac{G(p)}{\|G(p)\|}. \tag{2.1}$$

Proposition 2.6. *The map Ψ is a homeomorphism. It carries the product seam $\partial P_6 \times \partial P_6$ onto the Clifford torus*

$$T = \{(z_1, z_2) \in S^3 : |z_1| = |z_2| = 1/\sqrt{2}\}.$$

Proof. For $z \in S^3$, let $m(z) = \max(|z_1|, |z_2|)$. This number is positive, and $z/m(z)$ belongs to the product boundary. Direct substitution gives the inverse

$$\Psi^{-1}(z) = G^{-1}\left(\frac{z}{m(z)}\right).$$

Both maps are continuous. On the product seam, the two components of $G(p)$ both have norm one, so normalization gives two component norms $1/\sqrt{2}$. Conversely, equality of the component norms of z forces both components of $z/m(z)$ onto the unit circles. □

Remark 2.7. Raw radial normalization $p \mapsto p/\|p\|$ is also a homeomorphism $F \to S^3$, since every ray meets a convex boundary once. Along the seam its component radii $|u_1|, |u_2|$ generally differ, so it does not carry the seam onto $T_{\pi/4}$. The factorwise map $G = g \times g$ enforces equal component radii on the seam before the final normalization; this is the property used in Proposition 2.6.

The two halves of the boundary are

$$A = \partial P_6 \times P_6, \qquad B = P_6 \times \partial P_6, \qquad F = A \cup B.$$

Each is a solid torus, since $\partial P_6 \cong S^1$ and $P_6 \cong D^2$. Their common boundary is the seam. Under Ψ, the two halves become

$$\Psi(A) = \{|z_1| \geq |z_2|\}, \qquad \Psi(B) = \{|z_2| \geq |z_1|\}.$$

The first retracts to the core $(e^{i\alpha}, 0)$ and the second to $(0, e^{i\beta})$: contract the second component in the first region and normalize; the coordinate interchange gives the other.

Corollary 2.8 (Heegaard splitting)**.** *$F = A \cup B$ is a genus-one Heegaard splitting of the polyhedral three-sphere: two solid tori meeting along the seam $\partial P_6 \times \partial P_6$, carried by Ψ onto the standard splitting of S^3 along $T_{\pi/4}$ [13, §3.2], with the polygonal attaching data retained.*

2.3 Intrinsic distance and ambient interior

Throughout the monograph the round metric is transported through Ψ:

$$d_F(p, q) = \arccos\big(\Psi(p) \cdot \Psi(q)\big).$$

For state errors we use the Euclidean chord norm in the normalized embedding. If $z = \Psi(p)$ and $z' = \Psi(q)$, then

$$\|z - z'\| = 2\sin\left(\frac{d_F(p, q)}{2}\right).$$

Infinitesimal chord and round distances agree, and the global conversion is explicit. These distances differ from the piecewise flat length metric obtained by treating each prism facet as Euclidean. A great circle transported back to F can pass through facet interiors and need not follow a polyhedral edge.

Three uses of dimension are distinct. The configuration F has intrinsic dimension three: it is homeomorphic to S^3. Its ambient realization uses four coordinates. The body K that it bounds has four-dimensional interior. The boundary itself has empty interior as a subset of $\mathbb{R}^4$; every neighborhood of a boundary point also contains points outside the boundary. Equivalently for the normalized model, the equation $\|z\| = 1$ defines a three-dimensional boundary, while $\|z\| < 1$ defines the bounded open four-ball.

The statement that F has no four-dimensional interior concerns F as a subset of $\mathbb{R}^4$; F is the boundary of the four-dimensional filling K, whose interior points are not boundary configurations. Overlap in a projected image identifies observation fibers and attaches no facets to one another.

2.4 Orthogonal observation and its inverse

Write a normalized point as $z = (x, w) \in \mathbb{R}^3 \times \mathbb{R}$, and let $P(z) = x$. Solving the sphere equation gives

$$P(S^3) = \overline{B}^3, \qquad L_\sigma(x) = \left(x, \sigma\sqrt{1 - \|x\|^2}\right), \quad \sigma \in \{+1, -1\}. \tag{2.2}$$

Every interior point of the image has two representatives, and every point of its boundary sphere has one: the image of a three-sphere is a closed three-ball.

Definition 2.9 (Silhouette; sheet receipt)**.** The *silhouette* is the branch locus $\mathcal{B} = \{x \in \overline{B}^3 : \|x\| = 1\} = P(\{w = 0\})$, on which the differential of $P|_{S^3}$ drops rank from three to two. A *sheet receipt* is the sign σ of w, a receipt for P in the sense of Definition 1.1 on $S^3 \setminus \{w = 0\}$.

The topology is recovered by taking two copies of $\overline{B}^3$ and identifying their boundary spheres pointwise. The map induced by $(x, \sigma) \mapsto L_\sigma(x)$ is a continuous bijection onto S^3. The source quotient is compact and the sphere is Hausdorff, so it is a homeomorphism. The two signs distinguish the sheets over the open ball and coincide on the silhouette.

Proposition 2.10. *Projection is nonexpansive in chord norm. The pullback of the Euclidean metric of $\mathbb{R}^4$ under L_σ is*

$$g_{ij}(x) = \delta_{ij} + \frac{x_i x_j}{1 - \|x\|^2}$$

on the open ball B^3. On either fixed sheet, restricted to $\|x\| \le \sqrt{1 - a^2}$ with $0 < a \le 1$, the inverse lift is a^{-1}-Lipschitz; on $\overline{B}^3$ it is Hölder continuous of exponent $1/2$, and of no larger exponent.

Proof. Deleting a coordinate cannot increase Euclidean distance. For $\|x\| < 1$, differentiation of the lift gives

$$DL_\sigma(x)^T DL_\sigma(x) = I + \frac{xx^T}{1 - \|x\|^2}.$$

The derivative preserves lengths in directions orthogonal to x. In the radial direction its squared stretching factor is $(1 - \|x\|^2)^{-1}$. Hence its operator norm is at most $1/a$ on the stated ball. This ball is convex, so integration along the line segment between two observations proves the finite-distance bound. □

The margin a is a lower bound on the magnitude of the omitted coordinate. It quantifies the separation from the silhouette, where radial motion becomes difficult to reconstruct. The loss is intrinsic to this observation map. Taking $x_r = re_1$ and $x_1 = e_1$ on one sheet gives

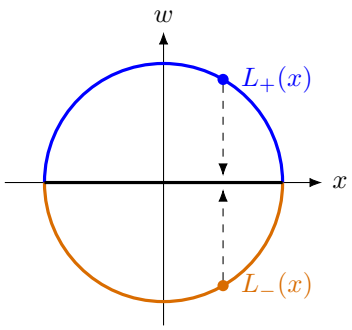
$$\|x_1 - x_r\| = 1 - r, \qquad \|L_\sigma(x_1) - L_\sigma(x_r)\| = \sqrt{2(1 - r)}.$$

Their ratio diverges as $r \uparrow 1$, so no uniform Lipschitz inverse exists on the closed ball. With $d = \|x - y\| \le 2$, the inequality $\left|\sqrt{s} - \sqrt{t}\right|^2 \le |s - t|$ yields

$$\|L_\sigma(x) - L_\sigma(y)\|^2 \le d^2 + 2d \le 4d.$$

This is the Hölder bound of exponent $1/2$; the radial example shows that the exponent is attained at the silhouette.

The normalized observation is the composite $P \circ \Psi$ on F. Direct linear projection of the original polytope boundary is a different map. For any projection deleting one coordinate, $P(F) = P(K)$: a



A two-dimensional section. In the full map, the interval is a filled B^3 and the endpoints are a silhouette S^2.

Figure 2.2: Orthogonal projection forgets the sheet. The representatives coincide on the silhouette, where the inverse changes from Lipschitz behavior on interior subballs to square-root behavior.

nonempty line section of the compact convex body has boundary endpoints, and those endpoints retain the same projected coordinates. Its image is the projection of the polytope, not the round ball obtained after normalization; an image is specified only together with the order of coordinate change and projection.

2.5 Stereographic coordinates and apparent scale

Stereographic projection from $N = (0, 0, 0, 1)$ is

$$\Sigma(x, w) = \frac{x}{1 - w}, \qquad \Sigma^{-1}(y) = \left(\frac{2y}{1 + \|y\|^2}, \frac{\|y\|^2 - 1}{1 + \|y\|^2} \right).$$

Substitution proves that these are inverse maps between $S^3 \setminus \{N\}$ and all of $\mathbb{R}^3$. The chart is particularly useful for displaying circles and their linking [13, §1]. It has no compact silhouette and forgets no sheet on its domain. Instead it omits one point of the sphere, whose approach corresponds to arbitrarily large chart coordinates.

Differentiating Σ^{-1} and taking the Euclidean inner product in $\mathbb{R}^4$ gives the round line element

$$ds^2 = \frac{4 \, \| \mathrm{d}y \|^2}{(1 + \|y\|^2)^2}.$$

The chart preserves angles, while a unit displacement in chart coordinates has round length $2/(1 + \|y\|^2)$. Conversely, its local magnification relative to round length is $(1 + \|y\|^2)/2 = 1/(1 - w)$. That factor becomes unbounded near the omitted pole.

Remark 2.11. An apparent enlargement in the chart is the conformal factor $(1 + \|y\|^2)/2$. Its identification with a change of physical distance requires an observable and a calibration of the chart against it. A chart displays each point and its neighborhood faithfully and their scale nonuniformly.

2.5.1 The volume of a whole and the volume of its image

The line element also determines how volume is represented. In three chart dimensions its conformal factor gives

$$\mathrm{d}V_{S^3} = \frac{8}{(1 + \|y\|^2)^3} \, \mathrm{d}y, \qquad \mathrm{Vol}(S^3) = 32\pi \int_0^\infty \frac{r^2}{(1 + r^2)^3} \, \mathrm{d}r = 2\pi^2.$$

For the integral, substitute $r = \tan t$; the remaining integral of $\sin^2 t \cos^2 t$ from zero to $\pi/2$ equals $\pi/16$. The unbounded chart therefore represents a sphere of finite intrinsic volume.

Orthogonal observation yields a complementary calculation. The graph metric in Proposition 2.10 has determinant $(1 - \|x\|^2)^{-1}$. Hence a single sheet has volume element

$$\mathrm{d}V_{\text{sheet}} = \frac{\mathrm{d}x}{\sqrt{1 - \|x\|^2}}, \qquad \operatorname{Vol}(\text{sheet}) = 4\pi \int_0^1 \frac{r^2}{\sqrt{1 - r^2}}\, \mathrm{d}r = \pi^2.$$

Here $r = \sin t$ reduces the integral to that of $\sin^2 t$. The two sheets recover total volume $2\pi^2$; their common silhouette has zero three-dimensional volume. The image ball itself has Euclidean volume $4\pi/3$. These quantities use different metrics and count different fibers.

More precisely, pushing the round volume through P gives density $2/\sqrt{1 - \|x\|^2}$ on the open ball. The density is unbounded at the silhouette but integrable. It records both the stretching of a sheet inverse and the multiplicity of the observation. The object is fixed; apparent scale and volume vary with the representation. A changing physical volume is a statement about a calibrated distance and observation model.

Chapter 3

Circle actions, winding, and algebraic representation

A motion of the sphere has a geometric, an algebraic, and a finite description. A permutation describes a sampled motion when an embedding intertwines the two; quaternion multiplication describes it when it yields the same coordinate map. The descriptions are identified through their common action on configurations.

Two families of great circles run along the diagonals of the angular square of the Clifford torus. Their winding vectors span $\mathbb{R}^2$ and an index-two sublattice of $\mathbb{Z}^2$. The index appears three times: as the antipodal fiber of the paired observation, as the cokernel of the phase matrix, and as the parity restriction of the finite clock.

3.1 The torus coordinates and two great-circle families

Use complex coordinates on the normalized sphere:

$$(z_1, z_2) = (\cos\eta\, e^{i\alpha}, \sin\eta\, e^{i\beta}), \qquad 0 \le \eta \le \pi/2.$$

For $0 < \eta < \pi/2$, each fixed η gives a torus T_η. The sphere metric in these coordinates is

$$ds^2 = d\eta^2 + \cos^2\eta\, d\alpha^2 + \sin^2\eta\, d\beta^2.$$

At $\eta = 0$ the phase β ceases to describe a nonzero coordinate; at $\eta = \pi/2$ the analogous statement holds for α. The two endpoint sets are consequently circles, the cores of the solid tori from Chapter 2. The Clifford torus is the intermediate level $T = T_{\pi/4}$.

Define two actions of $\mathbb{R}/(2\pi\mathbb{Z})$ by

$$C_+(t; z) = (e^{it} z_1, e^{it} z_2), \qquad C_-(t; z) = (e^{it} z_1, e^{-it} z_2).$$

Both preserve η, hence every intermediate torus. Each action is free: a fixed state under a phase step requires $e^{it} = 1$, because at least one complex coordinate is nonzero. Every orbit has period 2π.

Proposition 3.1. *The actions $C_\pm$ are the flows of the Killing vector fields $V_\pm(z) = (iz_1, \pm iz_2)$ of (S^3, g_{round}). Each orbit is a unit-speed great circle. On T, its coordinate winding is respectively $(1, 1)$ or $(1, -1)$.*

Proof. Set $v_+ = (iz_1, iz_2)$ and $v_- = (iz_1, -iz_2)$. In the real Euclidean inner product, each $v_\pm$ has norm one and is orthogonal to z. The corresponding orbit has the expression

$$C_\pm(t; z) = \cos t\, z + \sin t\, v_\pm.$$

It is therefore the unit circle in a real two-plane through the origin. Differentiating this expression gives speed one and $\frac{d}{dt}C_\pm(t; z) = V_\pm(C_\pm(t; z))$. Each $C_\pm(t)$ is unitary on $\mathbb{C}^2$, hence an isometry of the round sphere, so $V_\pm$ is Killing. The phase formulas increase (α, β) by (t, t) or $(t, -t)$, which gives the winding vectors after one full period. □

The coordinate curves with winding $(1, 0)$ and $(0, 1)$ also lie on T, but their radius in $\mathbb{R}^4$ is $1/\sqrt{2}$ and their affine planes do not pass through the origin. They are not great circles; the diagonal rulings are the great-circle rulings of T. The standard Clifford-torus geometry is discussed in [13, §3.2]. Every individual orbit has the topology of S^1; the distinction lies in its embedding and its action convention.

3.2 Hopf observations and their fibers

The observation constant along a circle action records its orbit rather than the phase within it. For our two conventions the Hopf maps are

$$H_+(z) = (2z_1\overline{z_2}, |z_1|^2 - |z_2|^2), \qquad H_-(z) = (2z_1 z_2, |z_1|^2 - |z_2|^2). \tag{3.1}$$

Their values lie in $S^2 \subset \mathbb{C} \times \mathbb{R}$, since

$$4\,|z_1|^2\,|z_2|^2 + (|z_1|^2 - |z_2|^2)^2 = (|z_1|^2 + |z_2|^2)^2 = 1.$$

In angular coordinates their complex components are $\sin(2\eta)e^{i(\alpha-\beta)}$ and $\sin(2\eta)e^{i(\alpha+\beta)}$, with common real component $\cos(2\eta)$.

Proposition 3.2. *The fibers of H_+ are exactly the C_+ orbits, and the fibers of H_- are exactly the C_- orbits, including the core circles.*

Proof. Substitution shows that each map is constant on the stated orbits. A nonpolar value determines the component magnitudes and hence a unique $\eta \in (0, \pi/2)$. For H_+ it also determines $\alpha - \beta$ modulo 2π. Any two pairs of phases with that difference differ by a common phase, so they are on one C_+ orbit. For H_- the fixed sum gives opposite phase increments and one C_- orbit. At either polar value, one complex coordinate vanishes and the other has arbitrary unit phase. Its fiber is the corresponding core circle. □

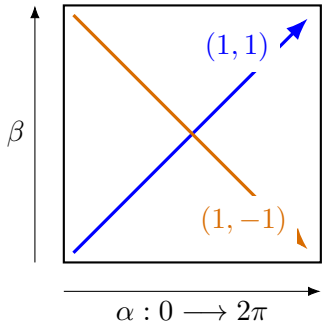
A Hopf observation discards a circle coordinate: it identifies an orbit and furnishes no representative of it. Lyons gives an elementary quaternion account of the Hopf construction [12]; here the phase formulas also prepare an explicit comparison of the two readings.

3.3 The paired reading and its antipodal ambiguity

Away from the two core circles, the pair $\mathcal{H} = (H_+, H_-)$ records η and the two phases

$$u = \alpha + \beta, \qquad v = \alpha - \beta \pmod{2\pi}.$$

The two base values have the same latitude $\cos(2\eta)$; $\mathcal{H}$ takes values in the diagonal-latitude subset of $S^2 \times S^2$.



Opposite sides are identified. The diagonals are great-circle rulings. Their determinant is -2: they span an index-two winding lattice.

Figure 3.1: An angular fundamental square for the Clifford torus. Opposite edges describe the same configurations, so the diagonal directions must be compared as integer windings as well as tangent directions.

Proposition 3.3. *For $z_1 z_2 \neq 0$, the fiber of $\mathcal{H}$ through z consists exactly of z and $-z$. A binary receipt recovers a representative after a phase-section convention has been fixed. At either core circle, the paired reading leaves a full phase undetermined.*

Proof. The phase transformation has matrix

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \det = -2.$$

On integer windings it gives the short exact sequence of abelian groups

$$0 \longrightarrow \mathbb{Z}^2 \xrightarrow{B} \mathbb{Z}^2 \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0, \tag{3.2}$$

whose cokernel is generated by the class of $(1,0) \equiv (0,1) \pmod{\operatorname{im} B}$. On the angular torus $\mathbb{R}^2/2\pi\mathbb{Z}^2$ the induced map is a twofold covering with deck group $\mathbb{Z}^2/\operatorname{im} B$: a phase increment in its kernel satisfies $\alpha + \beta = \alpha - \beta = 0$ modulo 2π, hence $\alpha = \beta$ and $2\alpha = 0$, so the kernel is $\{(0,0), (\pi,\pi)\}$, the two antipodal states.

For an explicit section, choose principal representatives $u, v \in [0, 2\pi)$ and set

$$z^0 = \left(\cos\eta\, e^{i(u+v)/2}, \sin\eta\, e^{i(u-v)/2}\right).$$

The retained sign $\epsilon \in \{+1, -1\}$ in $z = \epsilon z^0$ selects the state. At a core, both complex components of the paired output vanish; the remaining height is fixed and the nonzero state coordinate retains arbitrary phase. □

The principal-argument convention introduces cuts. Holding η and v fixed and increasing u by 2π carries a lifted state to its antipode while the paired observation returns to its initial value. No continuous global section of the angular covering exists (Section 4.2); a retained binary choice supplies a representative, and continuity of a reconstruction holds on a chosen phase neighborhood.

In the winding lattice the same determinant gives the index. Integer combinations of $(1,1)$ and $(1,-1)$ have coordinate windings of the same parity. Conversely, if (r,s) has that parity, then

$$(r,s) = \frac{r+s}{2}(1,1) + \frac{r-s}{2}(1,-1).$$

The two vectors span the index-two sublattice $\operatorname{im} B$ of $\mathbb{Z}^2$. They form a real basis of tangent directions and not an integer meridian–longitude basis; the phase ambiguity and the lattice index are the two ends of (3.2).

The paired map is quantitatively nondegenerate off the cores. Let the two S^2 outputs carry their Euclidean embedding norms and their product carry the sum-of-squares norm. Differentiating the angular formulas gives

$$\|D\mathcal{H}\ \mathrm{d}z\|^2 = 8\,d\eta^2 + 2\sin^2(2\eta)(d\alpha^2 + d\beta^2). \tag{3.3}$$

The cross terms from the sum and difference of phases cancel. Relative to the sphere metric, the squared singular values are

$$8, \qquad 8\sin^2\eta, \qquad 8\cos^2\eta.$$

The paired map has injective differential away from the cores. On a region where $\sin\eta, \cos\eta \geq b > 0$, its local inverse differential, on the image tangent space and for a chosen representative, has norm at most $1/(\sqrt{8}\,b)$. This is a differential bound within a local branch. It does not remove the antipodal ambiguity or justify crossing an argument cut. At the cores the bound degenerates, in agreement with the invisible phase in Proposition 3.3.

The antipodal receipt is distinct from the sheet receipt of orthogonal projection. The latter records the sign of a specified fourth coordinate. The former records which state lies above a paired phase value. They invert different observation maps, so agreement in the number of retained bits does not identify their contents.

3.4 Quaternion multiplication and the rotation group

Identify $\mathbb{R}^4$ with the quaternions by $q = z_1 + z_2 j$, with $ji = -ij$ and therefore $jz = \overline{z}\,j$ for $z \in \mathbb{C}$. Direct multiplication yields

$$e^{it}q = e^{it}z_1 + e^{it}z_2 j, \qquad qe^{it} = e^{it}z_1 + e^{-it}z_2 j.$$

Left multiplication realizes C_+ and right multiplication realizes C_-. Associativity gives their commutation, and multiplicativity of the quaternion norm gives preservation of the Euclidean state norm. In real coordinates their infinitesimal matrices are

$$\mathrm{diag}(J, J), \qquad \mathrm{diag}(J, -J), \qquad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Each rotates both coordinate planes; they are the Killing fields $V_\pm$ of Proposition 3.1.

The larger representation containing these circle actions is

$$\rho(a, b)q = aqb^{-1}, \qquad a, b \in S^3, \qquad \mathrm{Spin}(4) \cong Sp(1) \times Sp(1) \cong SU(2) \times SU(2),$$

acting on $\mathbb{H} \cong \mathbb{R}^4$, with $Sp(1)$ the unit quaternions. In this convention $C_+(t) = \rho(e^{it}, 1)$ and $C_-(t) = \rho(1, e^{-it})$. The representation is the usual double covering of $SO(4)$ [17, 4]. Its kernel can be verified directly. If $aqb^{-1} = q$ for every q, then $a = b$ by taking $q = 1$. This common quaternion must commute with every quaternion, hence is real; unit norm leaves $a = b = \pm 1$. Thus

$$\ker\rho = \{(1, 1), (-1, -1)\}.$$

The kernel is a central subgroup of order two of $\mathrm{Spin}(4)$, whose centre is $\{(\pm 1, \pm 1)\}$; it identifies two elements of the acting group that give one rotation in $SO(4)$. The fiber $\{z, -z\}$ of the paired Hopf observation $\mathcal{H} : S^3 \to S^2 \times S^2$ identifies two states that give one output. The two twofold structures lie in different spaces and act on different objects.

Our Clifford convention is $\mathrm{Cl}_{0,2} \cong \mathbb{H}$. Two abstract generators anticommute and each has square -1; their product supplies the third imaginary quaternion. For four ambient coordinates

the corresponding algebra is $\mathrm{Cl}_{0,4}$, with four anticommuting negative-square generators. Its even subalgebra is $\mathbb{H} \oplus \mathbb{H}$, providing the two quaternion factors associated with the spin representation [4]. The two-generator presentation describes a quaternion component; the four-generator algebra supplies the full ambient Clifford construction.

The quadratic form specifying the generator relations must be distinguished from the norm on configurations. The latter remains the positive Euclidean form $|z_1|^2 + |z_2|^2$. The negative generator squares specify an algebra convention; they do not themselves prescribe a physical time direction or a Lorentzian metric. Likewise, a Boolean sign schedule on a tape becomes an algebraic action only after a representation assigns its symbols to these operators. The representation is the data through which an operation acquires a geometric meaning.

There is an explicit isometry interchanging the selected families:

$$A(z_1, z_2) = (z_1, \overline{z_2}), \qquad AC_+(t)A^{-1} = C_-(t).$$

Complex conjugation reverses orientation in the second real plane, so A reverses orientation in $\mathbb{R}^4$. The choice of the two families includes an orientation convention, retained throughout.

3.5 Finite clocks and exact intertwining

Choose one orbit and an integer $N \geq 3$. Its uniform samples are

$$e_z(k) = C_+(2\pi k/N; z), \qquad k \in \mathbb{Z}/N\mathbb{Z}.$$

The clock shift $\sigma_j(k) = k + j$ satisfies

$$e_z(\sigma_j(k)) = C_+(2\pi j/N; e_z(k)). \tag{3.4}$$

The identity records the embedding, the time step, and the state on which the operation acts. It extends to linear representations. Let δ_k be the standard basis of $\mathbb{R}^N$, let $\Pi_j \delta_k = \delta_{k+j}$, and define $E_z \delta_k = e_z(k)$. Then

$$E_z \Pi_j = C_+(2\pi j/N) E_z.$$

The linear map E_z has a two-dimensional image, the orbit plane. Its restriction to $\{\delta_0, \ldots, \delta_{N-1}\}$ is injective, but its extension to all of $\mathbb{R}^N$ has a kernel of dimension $N - 2$. Exact agreement of the sampled operations does not imply an isomorphism of the ambient linear spaces.

The sampling error follows from the great-circle formula. For two phases,

$$\|C_+(t; z) - C_+(s; z)\| = 2\,|\sin((t - s)/2)|\,.$$

A nearest sample has phase error at most π/N, hence chord error at most $2\sin(\pi/(2N))$. The bound concerns one orbit; its base position under H_+ is still part of the configuration. An arbitrary permutation of clock labels may destroy cyclic adjacency; a reordered tape is read only with the permutation or its inverse retained.

3.6 The sixfold symmetry and its two sectors

For the polygonal construction, rotations through $\theta = \pi/3$ in either coordinate plane preserve the cells of K and F. Lemma 2.5 shows that g commutes with them, so Ψ intertwines the product

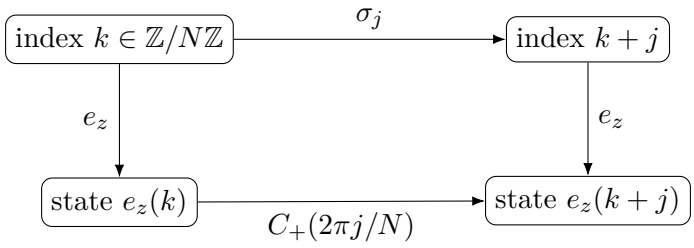


Figure 3.2: The clock permutation and the sampled continuous motion agree through a commuting diagram. The vertical maps specify which continuous states the clock labels represent.

symmetry $(\mathbb{Z}/6\mathbb{Z})^2$ with the corresponding round coordinate rotations. On the vertex labels the two selected great-circle steps are

$$S_+(j,k) = (j+1, k+1), \qquad S_-(j,k) = (j+1, k-1).$$

Each has order six, and they commute. Their generated translations have the form $(a+b, a-b)$ modulo six.

Proposition 3.4. *The subgroup $\langle S_+, S_- \rangle \subset (\mathbb{Z}/6\mathbb{Z})^2$ has index two and order 18. It has two orbits on the 36 vertex labels: the* even sector $j + k \equiv 0 \pmod 2$*, seeded by* $(0,0)$*, and the* odd sector $j + k \equiv 1 \pmod 2$*, seeded by* $(1,0)$*. The index is* $|\det B| = 2$*.*

Proof. The homomorphism $(a,b) \mapsto (a+b, a-b)$ on $(\mathbb{Z}/6\mathbb{Z})^2$ has kernel $\{(0,0),(3,3)\}$. Its image has $36/2 = 18$ elements, and every image label has even coordinate sum. There are exactly 18 such labels, so this necessary parity condition also characterizes the image. Translations by that subgroup have the subgroup and its odd-sum coset as their two orbits. □

On $(\mathbb{Z}/N\mathbb{Z})^2$ the kernel has order $\gcd(2, N)$: for odd N the diagonal steps generate all translations, and for even N an index-two subgroup. The two sectors are the finite realization at $N = 6$ of the cokernel of (3.2). A coordinate step $(j,k) \mapsto (j+1, k)$ exchanges them.

Any round action can be transported to F by conjugation with Ψ. It is then a homeomorphism preserving the transported round metric. Preservation of the polyhedral cells requires the additional discrete symmetry proved above. Both great-circle families have finite permutation descriptions and quaternion descriptions; the intertwining maps state the content of their agreement.

Chapter 4

Observation, cohomology, and retained information

The sphere of Chapter 2 is a complete state space; an observation is a function on it. A projected interior belongs to an image, a loop to a specified domain, and a lift to the information a map suppresses. This chapter treats recovery as an inverse problem on observation fibers, then constructs a two-cell complex on the separating torus and computes its cohomology with a specified inner product. The harmonic coordinates so obtained are the objects whose stability is proved in Chapter 5. The sphere identification rests on Ψ alone.

4.1 Enclosure and accessible domains

The homology of the whole enclosure is

$$H_1(S^3;\mathbb{Z}) = 0, \qquad H_3(S^3;\mathbb{Z}) = \mathbb{Z}.$$

One computation uses a CW structure with one zero-cell and one three-cell. Its cellular differentials vanish, giving these groups directly. Cellular homology and its agreement with singular homology are developed in Hatcher's second chapter [7, §2.2]. On the polyhedral boundary F, the consistently oriented sum of its three-dimensional facets is a representative of the top-dimensional generator. Adjacent facet boundaries cancel, and there are no four-cells in F that could make this cycle a cellular boundary. The homeomorphism Ψ carries its orientation class to the fundamental class of the sphere.

The separating torus has $H_1(T;\mathbb{Z}) = \mathbb{Z}^2$; the difference from $H_1(S^3;\mathbb{Z}) = 0$ is a change of domain. Exclude the two core circles

$$U = \{(e^{i\alpha}, 0)\}, \qquad V = \{(0, e^{i\beta})\}$$

are excluded. Their complement has global coordinates

$$S^3 \setminus (U \cup V) \cong T^2 \times (0, \pi/2), \qquad (\alpha, \beta, \eta) \longmapsto (\cos\eta\, e^{i\alpha}, \sin\eta\, e^{i\beta}).$$

For any fixed $\eta_0 \in (0, \pi/2)$, the homotopy replacing η by $(1-t)\eta + t\eta_0$ is a deformation retraction onto the torus at η_0. Both angular periods survive on this accessible region.

A coordinate loop on T nevertheless contracts in the full sphere. Holding β fixed while increasing η to $\pi/2$ shrinks the α loop to $(0, e^{i\beta})$. Holding α fixed while decreasing η to zero similarly contracts a β loop. The endpoint of either contraction belongs to an excluded core, so neither homotopy is

a contraction within the accessible region. The core complement is the domain restriction that preserves the two periods.

Linking gives a related statement. Two distinct Hopf fibers can be linked although either fiber, considered alone, is null-homotopic in S^3. The obstruction to contracting one while avoiding the other lies in a complement. An explicit account of the linked fibers appears in [12]. A statement about a cycle is a statement on a domain: the whole enclosure, its separating torus, a core complement, or a complement of selected fibers. These domains support different cycles within one construction.

4.2 Observation fibers and the recovery problem

Definition 4.1. An *exact observation* on a state domain D is a map $O : D \to Y$, with fiber $O^{-1}(y)$ at y. A *receipt* for O is a map $R : D \to A$ with $(O, R) : D \to Y \times A$ injective; this is Definition 1.1 with E the identity.

A receipt may be stored in a larger configuration, a finite register, or a record of operations; its implementations differ in cost and agree in the states their values distinguish.

An observation determines the equivalence relation

$$s \sim_O t \quad \Longleftrightarrow \quad O(s) = O(t).$$

Let $\pi : D \to D/\!\sim_O$ be the quotient map. As sets, the quotient and $O(D)$ are in bijection through $[s] \mapsto O(s)$.

Proposition 4.2 (Descent and recovery)**.** *A reader $q : D \to Z$ factors through π if and only if q is constant on each observation fiber. Exact recovery of every state from the pair $(O(s), R(s))$ is possible if and only if R is injective on each fiber of O.*

Proof. If $q = \bar{q} \circ \pi$, equal classes have equal readings. Conversely, constancy on each class makes $\bar{q}([s]) = q(s)$ well-defined, and this is the required factorization. For the second assertion, injectivity on every fiber is equivalent to injectivity of (O, R). An injective map has an inverse on its image, while any proposed recovery map would force injectivity by recovering equal outputs to equal states. □

Remark 4.3. Proposition 4.2 is algebraic and establishes no measured fact: which state is present is determined only once an observation value and its receipt are supplied. Deduction ends and observation begins at that argument. Continuity and quantitative inverse estimates require further hypotheses. A continuous injective map from a compact state space to a Hausdorff output space is a homeomorphism onto its image. A discontinuously chosen lifting label need not satisfy that hypothesis. In the orthogonal projection, the sheet choice separates the two lifts away from the silhouette, while the two lifts coincide at the silhouette itself. A stable continuous inverse is therefore stated on a fixed sheet with a positive margin, as in Proposition 2.10.

A discrete ambiguity persists in the presence of local inverse formulas. Away from the cores, the observable angles are $u = \alpha + \beta$ and $v = \alpha - \beta$ modulo 2π. The corresponding torus map has integer matrix

$$B = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \det B = -2.$$

Its fibers have two points, and there is no continuous global section: a continuous section would induce an integer matrix C on the fundamental groups of the tori satisfying $BC = I$. Taking determinants would give $-2 \det C = 1$, which is impossible over $\mathbb{Z}$. Local choices can be made on

Observation	Identified information	Retained lifting datum
Orthogonal P	Two fourth-coordinate signs	Sheet, except at the silhouette
Paired H_+, H_-	z and $-z$ off the cores	Antipodal bit; full phase at a core
Spin action ρ	(a, b) and $(-a, -b)$	Lift of a rotation to the acting group
Scalar equation $r(y) = 0$	Choice among solution roots	Selected interval or root branch
Constant quotient	Every admitted state	A retained representative or its code

Table 4.1: Different observation maps give different lifting problems. Equal fiber cardinalities alone do not identify their state spaces or inverses.

angle charts, but crossing their cuts requires the retained choice described in Chapter 3. At a core, an entire phase disappears, so a binary label no longer suffices.

The other rows of Table 4.1 involve independent choices. For example, $13y^2 - 36y + 18 = 0$ has roots $(18 \pm 3\sqrt{10})/13$. Selecting a root fixes a scalar branch and specifies no sign of w in (2.2) until an encoding connects the scalar variable with that coordinate. Likewise, the kernel of the spin action concerns a representation of an acting group, rather than the antipodal ambiguity of a state observation.

A quotient generated by a total order is the extreme case. If every comparable pair is identified, then every pair of elements is identified, because a total order makes every pair comparable. The resulting quotient has one point, and every descended numerical reader is constant. A larger output may retain an integer alongside that quotient. Recovery then uses this retained integer; it cannot be attributed to distinctions within the one-point quotient.

4.3 Finite ambiguity and approximate readings

Exact fibers are distinct from numerical neighborhoods. For $\varepsilon > 0$, the relation $\|O(s) - O(t)\| \le \varepsilon$ need not be transitive: the real observations $0, \varepsilon, 2\varepsilon$ already provide a counterexample. An error ball describes the states compatible with a bounded discrepancy. It becomes an equivalence class only under a further construction.

Proposition 4.4 (Size of retained information)**.** *If an observation fiber contains m admitted states, any receipt alphabet that supports exact recovery on that fiber has at least m symbols. A fixed-length binary encoding requires at least $\lceil \log_2 m \rceil$ bits in that worst case.*

Proof. An alphabet with fewer than m symbols assigns the same receipt to two states in the fiber. Their observations are already equal, so the combined observation and receipt cannot be injective. A binary word of length b has 2^b possible values; hence $2^b \ge m$. □

For a uniform random state S on a finite set of size N, let m_y be the size of the fiber at observation y. The conditional distribution on that fiber is uniform. Its conditional entropy, measured in bits, is therefore

$$H(S \mid O(S)) = \sum_{y \in O(D)} \frac{m_y}{N} \log_2 m_y.$$

For the 36-state construction of Chapter 6, the orthogonal reader has twelve singleton fibers and twelve double fibers. Its conditional entropy is $24/36 = 2/3$ bit. The paired Hopf reader has eighteen double fibers and conditional entropy one bit. A constant reader leaves $\log_2 36$ bits. The worst-case alphabet bound and the average information answer different questions; the latter depends on the input distribution. Neither assigns an energy cost to recording or erasing information.

4.4 A cellular model of the accessible torus

Definition 4.5 (The torus complex)**.** X_{torus} is the CW complex obtained from the unit square with opposite sides identified and divided by the diagonal from $(0,0)$ to $(1,1)$: one zero-cell, oriented one-cells a, b (horizontal, vertical) and c (diagonal), and two-cells f_1, f_2 attached along the words abc^{-1} and $ca^{-1}b^{-1}$.

Cellular chains over $\mathbb{R}$ give

$$\partial a = \partial b = \partial c = 0, \qquad \partial f_1 = a + b - c, \qquad \partial f_2 = -a - b + c.$$

The two face boundaries are negatives. Their sum is the fundamental two-cycle of the torus. The image of ∂_2 is one-dimensional, so

$$H_1(T;\mathbb{R}) = \mathbb{R}^3/\operatorname{span}\{a+b-c\}, \qquad H_2(T;\mathbb{R}) = \operatorname{span}\{f_1+f_2\}.$$

The diagonal represents $a + b$ in first homology. The torus topology is that of the identified square.

Choose the cell cochains dual to these oriented cells and equip their coordinate spaces with standard Euclidean inner products. The cochain complex is

$$C^0 = \mathbb{R} \xrightarrow{d_0=0} C^1 = \mathbb{R}^3 \xrightarrow{d_1 = \begin{pmatrix} 1 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix}} C^2 = \mathbb{R}^2. \tag{4.1}$$

A cochain $\alpha = (x, y, z)$ assigns the three periods x, y, z to the oriented loops. Closedness is exactly $x+y-z = 0$. A raw record may fail it; when it fails, the constraint is enforced exactly, by orthogonal projection onto $\ker d_1$, not approached through a penalty term, and the harmonic representative selected below is obtained by that exact projection rather than by a relaxation parameter. Since $d_0 = 0$, no nonzero one-cochain is exact, and every cohomology class has one representative in $\ker d_1$. Thus

$$H^1(T;\mathbb{R}) = \{(A, B, A+B) : A, B \in \mathbb{R}\}.$$

The integral periods form the lattice $A, B \in \mathbb{Z}$. The diagonal contributes a consistency relation, not a third independent period.

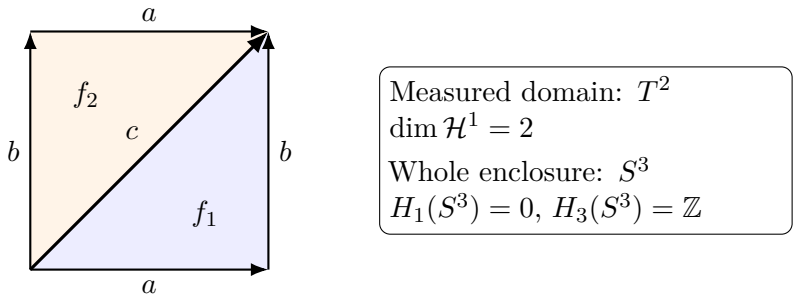


Figure 4.1: Opposite edges and all four corners are identified. The accessible torus has its own period domain within the whole enclosure.

4.5 Harmonic representatives and measured periods

For a finite cochain complex with inner products, the degree-one Hodge operator is $\Delta_1 = d_0 d_0^* + d_1^* d_1$. Its kernel consists of closed cochains orthogonal to exact cochains. Indeed,

$$\langle \Delta_1 \alpha, \alpha \rangle = \|d_0^* \alpha\|^2 + \|d_1 \alpha\|^2 .$$

Moreover $\ker d_1 = \operatorname{im} d_0 \oplus \mathcal{H}^1$ orthogonally: project a closed cochain onto $\operatorname{im} d_0$, and the remaining component is still closed and annihilated by d_0^*. Consequently every cohomology class has a unique harmonic representative. This finite-dimensional form of Hodge theory underlies combinatorial inference [9].

Proposition 4.6 (Spectrum and harmonic response of the torus complex)**.** *With the Euclidean inner product on $C^1 \cong \mathbb{R}^3$ and $v = (1,1,-1)^T$,*

$$\Delta_1 = d_1^* d_1 = 2vv^T = 6P_v, \qquad P_v = \tfrac{1}{3} vv^T, \qquad \mathcal{H}^1 = v^\perp = \{(A, B, A+B) : A, B \in \mathbb{R}\}. \tag{4.2}$$

P_v is the orthogonal projection onto $\operatorname{span}(v) = \mathcal{H}^{1\perp}$*, $\sigma(\Delta_1) = \{0, 0, 6\}$, and the spectral gap is $\lambda = 6$. For the period map $M : C^1 \to \mathbb{R}^2$, $M(x,y,z) = (x,y)$,*

$$\gamma = \min_{h \in \mathcal{H}^1,\, \|h\|=1} \|Mh\| = \frac{1}{\sqrt{3}}.$$

Proof. Both rows of d_1 are $\pm v^T$, so $d_1^* d_1 = 2vv^T$, and $\|v\|^2 = 3$ gives $2vv^T = 6P_v$. Since $d_0 = 0$, $\Delta_1 = d_1^* d_1$, which vanishes on $v^\perp$ and sends v to $6v$. For a unit $h = (x,y,z) \in v^\perp$, the Rayleigh quotient is $\|Mh\|^2 = 1 - z^2$. The maximum of $z^2 = \langle h, e_3 \rangle^2$ over unit $h \in v^\perp$ is $\|e_3 - P_v e_3\|^2 = 1 - \frac{1}{3} = \frac{2}{3}$, attained at $h = (1,1,2)/\sqrt{6}$. Hence $\gamma^2 = \frac{1}{3}$. □

The dimension of cohomology follows from incidence; the spectral value 6 and the response $1/\sqrt{3}$ also use the inner products. The restriction $M|_{\mathcal{H}^1}$ has inverse $(A,B) \mapsto (A, B, A+B)$ of norm exactly $\sqrt{3} = 1/\gamma$, attained at $A = B$. One period is not injective on a two-dimensional harmonic space; two periods determine the harmonic representative without changing the complex.

Removing both faces while retaining the three loop edges makes d_1 vanish and increases the harmonic dimension to three. Attaching a disk along a instead makes that cycle a boundary. Both operations alter the cellular model. A small cochain coefficient on an unchanged complex is information about a cochain and attaches no cell.

4.6 Transport of representation and metric

A change of coordinates preserves the preceding calculations when it carries both incidence and inner product. Let $S_k : C^k \to C'^k$ be invertible maps satisfying

$$d'_k S_k = S_{k+1} d_k, \qquad \langle S_k u, S_k v \rangle' = \langle u, v \rangle .$$

The second condition makes S_k an isometry for the transported inner products. Taking adjoints of the first gives

$$(d'_k)^* S_{k+1} = S_k d_k^*, \qquad \Delta'_1 S_1 = S_1 \Delta_1.$$

Hence harmonic representatives and the complete spectrum are transported together. If the period reader is transported as $M' = M S_1^{-1}$ and its output norm is fixed, its smallest harmonic response is also unchanged.

An incidence-preserving isomorphism without metric transport still identifies the cohomology groups. It need not preserve adjoints, harmonic representatives as coordinate vectors, or positive eigenvalues. A change of cell weights is a change of the calibrated geometry of the finite problem. Compatible maps and controlled inner products play a central role in finite element exterior calculus [1, §3]. Their finite-dimensional form separates the topological statement that two periods exist from the quantitative statement that they are recovered with constant $1/\gamma$.

Chapter 5

Stability of observation and reconstruction

Exact recovery is the question whether an output determines a state; stable recovery is the bound on the change of state under a change of output. The first is settled by fibers, receipts, and incidence; the second by norms and positive margins. An injective map may have a poorly conditioned inverse, and an accurate observation may leave a fiber unresolved.

The exact choices being fixed, geometry supplies the inverse bound on a projection sheet; the cochain complex supplies a residual estimate transverse to its harmonic space; measured periods supply control within that space; a retained word of operations bounds accumulated computational error. The estimates meet in a comparison of two representations of one calibrated observable.

5.1 Residual energy and the harmonic component

Let

$$C^0 \xrightarrow{d_0} C^1 \xrightarrow{d_1} C^2, \qquad d_1 d_0 = 0,$$

be a finite real cochain complex. Equip each space with a positive definite inner product, and form the corresponding adjoints. Throughout this section all norms are induced by those fixed inner products. Define

$$\Delta_1 = d_0 d_0^* + d_1^* d_1, \qquad \mathcal{H}^1 = \ker \Delta_1.$$

The energy identity

$$\langle \Delta_1 \alpha, \alpha \rangle = \|d_0^* \alpha\|^2 + \|d_1 \alpha\|^2$$

shows that $\mathcal{H}^1 = \ker d_0^* \cap \ker d_1$. The first residual measures failure to be orthogonal to exact cochains, and the second measures failure of closedness. Neither residual detects a harmonic component. Finite Hodge theory gives the appropriate orthogonal separation [9, 1].

Write $\alpha = h + e$, where h is the orthogonal projection of α onto $\mathcal{H}^1$ and $e \perp \mathcal{H}^1$. When the orthogonal complement is nonzero, its smallest eigenvalue under Δ_1 is positive. Denote this eigenvalue by λ. If the complement is zero, every cochain is harmonic and the transverse estimate is unnecessary.

Proposition 5.1 (Residual estimate)**.** *Suppose $\mathcal{H}^{1\perp} \neq \{0\}$ and the least eigenvalue of Δ_1 on this space is $\lambda > 0$. Then*

$$\|e\| \leq \lambda^{-1/2} \left(\|d_0^* \alpha\|^2 + \|d_1 \alpha\|^2 \right)^{1/2}.$$

Equality is possible when e belongs to the eigenspace of λ.

Proof. Expand e in an orthonormal eigenbasis of the self-adjoint operator Δ_1. Every eigenvalue on the orthogonal complement is at least λ, so $\lambda \|e\|^2 \le \langle \Delta_1 e, e\rangle$. The harmonic component contributes zero to both residuals. The energy identity then gives the estimate, and an eigenvector at λ gives equality. □

The estimate controls distance to the harmonic space. It cannot determine which harmonic representative is present. In particular, adding any $h_0 \in \mathcal{H}^1$ leaves both residuals unchanged. Period measurements are required by this kernel; no decrease of a residual tolerance removes it.

5.2 Period measurements and uniqueness

Let $M : C^1 \to \mathbb{R}^p$ be a linear period reader, with a specified Euclidean norm on its output. Suppose $\mathcal{H}^1$ is nonzero. Define the smallest harmonic response by

$$\gamma = \min_{h \in \mathcal{H}^1,\, \|h\|=1} \|Mh\| .$$

Compactness of the unit sphere shows that the minimum is attained. It is positive exactly when $M|_{\mathcal{H}^1}$ is injective. The constant γ is the quantitative form of uniqueness.

Proposition 5.2 (Recovery from periods)**.** *Assume $\gamma > 0$. For data $m = Mh + \xi$ with $h \in \mathcal{H}^1$ and $\|\xi\| \le \nu$, the unique least-squares estimate on $\mathcal{H}^1$ satisfies*

$$\left\|\widehat{h} - h\right\| \le \nu/\gamma. \tag{5.1}$$

If the same measurement protocol instead supplies $m = M\alpha + \xi$ for $\alpha = h + e$, the estimate of its harmonic component obeys

$$\left\|\widehat{h} - h\right\| \le \frac{\nu + \|M\|\,\|e\|}{\gamma}.$$

Proof. Choose orthonormal coordinates on $\mathcal{H}^1$. The restriction $B = M|_{\mathcal{H}^1}$ has full column rank and least singular value γ. Its least-squares inverse is its Moore–Penrose pseudoinverse $B^\dagger$, with norm $1/\gamma$. Hence $\widehat{h} - h = B^\dagger \xi$. In the second protocol, the effective data error is $Me + \xi$; apply the triangle inequality before the same inverse bound. □

The two protocols are two orders of operation: removal of the transverse component before measurement of periods, or measurement of the uncorrected cochain with its contribution carried in the data. Singular-value methods provide the general linear-algebraic setting for these statements [6, Chapter 5].

The residual and the period reader can also be treated as a single observation. Give the output below the product inner product and define

$$\mathcal{A}\alpha = (d_0^*\alpha, d_1\alpha, M\alpha).$$

Theorem 5.3 (Complete linear observation)**.** *The operator $\mathcal{A}$ is injective if and only if M is injective on $\mathcal{H}^1$. When this condition holds and $C^1 \neq \{0\}$, its least singular value τ is positive. For any two cochains,*

$$\|\alpha - \beta\| \le \frac{1}{\tau} \|\mathcal{A}\alpha - \mathcal{A}\beta\| .$$

Least-squares reconstruction from data with error at most $\nu_{\mathcal{A}}$ has state error at most $\nu_{\mathcal{A}}/\tau$.

Proof. The first two components of $\mathcal{A}\alpha = 0$ imply $\alpha \in \mathcal{H}^1$. Its remaining component then vanishes only at zero precisely when $M|_{\mathcal{H}^1}$ is injective. In finite dimensions, injectivity gives a positive least singular value. Apply its defining inequality to $\alpha - \beta$, or apply the corresponding pseudoinverse to the data error. □

A numerical value for τ presupposes the output norm, including any weights between residuals and periods. Rescaling one output component changes the singular values of the combined observation without changing its kernel. Such scaling is a calibration and is part of the definition of the reader.

5.3 The torus calculation in full

For the torus of (4.1), $d_0 = 0$, $\lambda = 6$, and $\gamma = 1/\sqrt{3}$. If $\alpha = (x, y, z)$ and $\rho = x + y - z$, the orthogonal decomposition is

$$e = \frac{\rho}{3}(1, 1, -1), \qquad h = \alpha - e, \qquad \|e\| = \frac{|\rho|}{\sqrt{3}} = \frac{\|d_1\alpha\|}{\sqrt{6}}.$$

Every transverse component lies in the sole positive eigenspace, so the residual bound is attained. Corrected period data (m_a, m_b) recover

$$\widehat{h} = (m_a, m_b, m_a + m_b).$$

The reconstruction operator has norm $\sqrt{3}$, as calculated in the preceding chapter. Errors with equal components in the two periods attain this amplification.

The combined observation also has an exact singular-value calculation. With the Euclidean product norm and $M(x, y, z) = (x, y)$,

$$\mathcal{A}^*\mathcal{A} = \begin{pmatrix} 3 & 2 & -2 \\ 2 & 3 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

The vector $(1, -1, 0)$ is an eigenvector with eigenvalue 1. On the span of $(1, 1, 0)/\sqrt{2}$ and $(0, 0, 1)$, the matrix is

$$\begin{pmatrix} 5 & -2\sqrt{2} \\ -2\sqrt{2} & 2 \end{pmatrix},$$

whose characteristic polynomial is $t^2 - 7t + 2$.

Theorem 5.4 (Closed-form inverse for the torus)**.** *The joint observation $\mathcal{A} = (d_1, M) : C^1 \to \mathbb{R}^2 \oplus \mathbb{R}^2$ of the torus complex has least singular value*

$$\tau = \sqrt{\frac{7 - \sqrt{41}}{2}} > 0,$$

and $\|\alpha - \beta\| \le \tau^{-1} \|\mathcal{A}\alpha - \mathcal{A}\beta\|$ for all cochains.

Proof. The eigenvalues of $\mathcal{A}^*\mathcal{A}$ are 1 and $(7 \pm \sqrt{41})/2$; the least is $(7 - \sqrt{41})/2$. Apply Theorem 5.3. □

The constants λ and γ are the two sources of control; τ combines them in one data norm.

5.4 Perturbations and calibration margins

A positive inverse margin can survive a bounded change of the reader. Let $\widetilde{M}$ be another period map on the same spaces, and suppose $\left\|\widetilde{M} - M\right\| \le \zeta < \gamma$. Then, for every $h \in \mathcal{H}^1$,

$$\left\|\widetilde{M}h\right\| \ge \|Mh\| - \left\|(\widetilde{M} - M)h\right\| \ge (\gamma - \zeta)\,\|h\|\,.$$

The perturbed restriction remains injective, and its inverse norm is at most $(\gamma - \zeta)^{-1}$. The assumption fixes the harmonic space and both norms; it compares operators on the same calibrated problem.

If data were generated by M but reconstruction uses $\widetilde{M}$, the calibration discrepancy contributes a bias. For $m = Mh + \xi$, least-squares reconstruction using $\widetilde{M}|_{\mathcal{H}^1}$ satisfies

$$\left\|\widehat{h} - h\right\| \le \frac{\nu + \zeta\,\|h\|}{\gamma - \zeta}.$$

A uniform bias allowance requires in addition a bound on admitted state amplitudes, which invertibility of the perturbed reader does not supply.

For a self-adjoint positive semidefinite operator $\widetilde{\Delta}_1$ on the same inner-product space, assume that

$$\ker \widetilde{\Delta}_1 = \mathcal{H}^1, \qquad \left\|\widetilde{\Delta}_1 - \Delta_1\right\| \le \beta < \lambda.$$

On the common orthogonal complement, the Rayleigh quotient gives

$$\left\langle \widetilde{\Delta}_1 e, e \right\rangle \ge (\lambda - \beta)\,\|e\|^2\,.$$

Its gap is at least $\lambda - \beta$. The shared-kernel hypothesis is essential to this particular estimate. A different attachment map can alter the harmonic space, while a different cell metric can change which coordinate vectors represent it. Such changes require an identification of spaces and metrics before an operator perturbation is defined.

For the explicit torus, a period-map perturbation of at most $1/100$ leaves response at least $1/\sqrt{3} - 1/100 > 0$. A same-kernel Hodge-operator perturbation of at most $1/10$ leaves a gap of at least $59/10$. These margins are stated in the specified operator norms for the fixed inverse problem; a new cell attachment is not a calibration error.

5.5 Composition of operations and the role of order

Let an exact word of operations be $T_n \circ \cdots \circ T_1$. Suppose T_j is L_j-Lipschitz on a domain containing both the exact and computed inputs. Let the computed step have error at most ε_j relative to T_j applied to that computed input. If E_j is the state error after the jth step, then

$$E_j \le L_j E_{j-1} + \varepsilon_j.$$

Successive substitution proves

$$E_n \le E_0 \prod_{j=1}^{n} L_j + \sum_{j=1}^{n} \varepsilon_j \prod_{k=j+1}^{n} L_k. \tag{5.2}$$

The products describe how later operations amplify an earlier error. Round quaternion actions are isometries for the chord metric, so their L_j equal one and the bound reduces to addition of step

errors. Quantization can be included as a bounded step error when its rule and norm are stated. A different generator or a lost inverse permutation changes the exact word being compared and needs a corresponding change in the observation model.

Order is the comparison of two composites on one state domain. For unit vectors e, f in a real inner-product space, put

$$Q_e = I - ee^T, \qquad Q_f = I - ff^T, \qquad c = \langle e, f \rangle .$$

Multiplication gives

$$[Q_e, Q_f] = c(ef^T - fe^T), \qquad \|[Q_e, Q_f]\| = |c| \sqrt{1 - c^2} \le \tfrac{1}{2}.$$

For $|c| < 1$, choose $g = (f - ce)/\sqrt{1 - c^2}$. On the orthonormal basis e, g, the commutator has matrix

$$\begin{pmatrix} 0 & c\sqrt{1 - c^2} \\ -c\sqrt{1 - c^2} & 0 \end{pmatrix};$$

it vanishes on the orthogonal complement. The endpoint cases $|c| = 1$ give zero directly. The inequality follows from $c^2(1 - c^2) \le 1/4$. With the convention $(a \wedge b)x = a \langle b, x \rangle - b \langle a, x \rangle$ identifying $\bigwedge^2 \mathbb{R}^n$ with skew operators, $ef^T - fe^T$ is the operator of the bivector $e \wedge f$, of norm $\|e \wedge f\| = \sqrt{1 - c^2}$. Hence

$$[Q_e, Q_f] = c\,(e \wedge f), \qquad \|[Q_e, Q_f]\| = |\langle e, f \rangle|\ \|e \wedge f\| :$$

the product of the inner and exterior parts of the Clifford product $ef = \langle e, f \rangle + e \wedge f$.

Its exponential is a rotation supported on the plane of $e \wedge f$, with angular speed $|c| \sqrt{1 - c^2}$. The projections themselves are noninvertible. Moreover, the commutator has rank at most two, whereas the generators $\operatorname{diag}(J, \pm J)$ of nontrivial Hopf motion have rank four. Its relation to a Hopf action requires a comparison map; exponentiation of an order defect does not identify the two actions.

5.6 Comparison through a common observable

The preceding estimates concern different spaces and are compared through the observable they recover. Let $q : D \to Y$ be a dimensionless calibrated observable in a normed space. For representation i, let $E_i : D \to Z_i$ be an encoding and $r_i : Z_i \to Y$ its reader. Suppose

$$\|r_i(E_i(s)) - q(s)\| \le \eta_i, \qquad \|\widehat{z}_i - E_i(s)\| \le \delta_i,$$

where η_i bounds the exact diagram defect and δ_i bounds the state error. Assume that r_i is L_i-Lipschitz throughout the relevant recovery domain.

Proposition 5.5 (Comparison of representations)**.** *Under these assumptions, observations of the same admitted state satisfy*

$$\|r_1(\widehat{z}_1) - r_2(\widehat{z}_2)\| \le \eta_1 + \eta_2 + L_1\delta_1 + L_2\delta_2. \tag{5.3}$$

Proof. Insert $r_1(E_1(s))$, then $q(s)$, then $r_2(E_2(s))$ between the two computed readings. The triangle inequality bounds the four successive differences, and the reader Lipschitz constants bound the two differences arising from state error. □

The comparison requires one specified q on one admitted state domain. Similar formulas in different representations do not establish this common observable. The diagram defects record the discrepancy remaining after the encodings and readers have been chosen. Conversely, agreement of two computed readings supplies no inverse statement unless the common observation, together with its retained data, distinguishes the states in question.

Structure	Exact data fixed	Stability constant or bound
Geometry	Sheet and inverse domain	$\delta x/a$ for lifted state error
Algebra	Generator word and action map	Equation (5.2)
Topology	Incidences and period domain	$\lambda^{-1/2}$ residual, γ^{-1} period error
Common reading	Calibrated observable and readers	Equation (5.3)

Table 5.1: The inverse estimates apply after the discrete lifting choices and the exact domains have been specified.

5.7 Trace errors as inputs to reconstruction

A trace observable supplies a bound that enters this comparison. Under the earlier instrument's uniform-grid sampling, boundary, knot-noise, and Hermite-slope hypotheses, its estimate is

$$\|u - f\|_\infty \leq \frac{7}{4}\Omega_f(h) + \frac{5}{2}\epsilon + \frac{h^2}{4\sqrt{6}}\|r\|_2, \qquad \Omega_f(h) \geq \omega_f(h).$$

Here h is the grid spacing, ϵ bounds knot errors, $\|r\|_2$ is the Euclidean norm of the slope residual, and ω_f is the modulus of continuity. The first term controls unresolved variation of the target, the second controls the admitted knot perturbations, and the last controls the error left by the slope calculation. The precise construction is the complete error budget of the earlier instrument study; Appendix A records its location. Classical minimum properties of splines supply the variational context [3, 2].

Using that estimate in a sphere-valued reconstruction requires the coordinate reader and its conversion to the sphere metric. A scalar trace error may be amplified by a subsequent inverse, especially near a projection silhouette. Its contribution then enters through the appropriate reader constant in (5.3). Coordinatewise accuracy does not establish a cell attachment, select an omitted sheet, or determine an unmeasured harmonic period.

All bounds in this chapter are deterministic. Statistical uncertainty requires an accompanying probability model and a stated coverage interpretation. A norm inequality alone supplies no confidence level; the distinction between a measurement model, uncertainty propagation, and coverage is treated in the metrological references [10, 11].

The inverse estimates have exact limits. The two points (x, w) and $(x, -w)$ have identical orthogonal observations but are separated by chord distance $2|w|$. Removing both certifying faces of the torus complex introduces an additional harmonic direction with zero local residual. These failures occur before any small numerical tolerance is considered. Once the domain, incidence structure, and retained lifting information are fixed, the positive constants proved above quantify reconstruction within that mathematical model.

Chapter 6

A finite realization and its reconstruction

The finite realization has every state, observation fiber, and recovery map in closed form. The product polytope supplies its vertices, the two circle actions its operations, and the torus complex a two-period record. Quantitative reconstruction depends on three data: separation of the finite observations, a receipt within each fiber, and the response of the period map.

The realization uses two six-position clocks, matching the hexagonal factors. It is a finite sector of the continuous enclosure. Its theorems establish correspondence among specified descriptions of that sector; interpretation of a physical apparatus requires further identifications, and none is made here.

6.1 States, encodings, and the action of the clocks

Let

$$D = (\mathbb{Z}/6\mathbb{Z})^2, \qquad \theta = \pi/3,$$

and choose representatives $0 \le j, k < 6$ whenever an integer is printed. This is the rule of admission for the finite record: a state enters it only as a pair of residues with these canonical representatives, and every later statement about a state refers to the pair so admitted. Recall the vertices $u_j = (\cos j\theta, \sin j\theta)$ of the hexagon. The two geometric encodings are

$$E_F(j,k) = (u_j, u_k) \in F, \qquad E(j,k) = \Psi(E_F(j,k)) = \frac{1}{\sqrt{2}}(e^{ij\theta}, e^{ik\theta}). \tag{6.1}$$

At a polygon vertex the radial map of Chapter 2 reaches the unit circle. Consequently the displayed formula follows directly from the definition of Ψ. All 36 states lie on the separating Clifford torus, and their preimages are vertices of the product polytope.

Both encodings are injective. A sixth root of unity uniquely determines its exponent modulo six, so reading the two complex coordinates gives an inverse on $E(D)$. An alternative encoding uses the integer

$$n = j + 6k \in \{0, \dots, 35\}.$$

Division with remainder gives $j = n \bmod 6$ and $k = \lfloor n/6 \rfloor$. These are different codomains for the same admitted states, with explicit maps between their images. Their equivalence concerns the retained information; it does not require the ambient integer interval to possess the geometry of the torus.

Let T_1 increment the first index and T_2 the second. Their actions on (6.1) multiply the corresponding complex coordinate by $e^{i\theta}$. The selected simultaneous steps are

$$S_+ = T_1T_2, \qquad S_- = T_1T_2^{-1}.$$

They therefore agree exactly with the parameter-θ restrictions of the two round circle actions of Chapter 3. The agreement is an intertwining identity between a permutation action and a geometric action.

The selected steps preserve $j + k$ modulo two. More explicitly, applying $S_+^a S_-^b$ changes the two indices by

$$(a + b, a - b) \pmod 6.$$

The kernel of this homomorphism from $(\mathbb{Z}/6\mathbb{Z})^2$ consists of $(0, 0)$ and $(3, 3)$. Its image has 18 elements and lies in the 18-element set of increments with even coordinate sum, hence equals that set. There are exactly two orbits: seed $(0, 0)$ generates the even sector, and seed $(1, 0)$ generates the odd sector.

A seed and a retained operation word determine the resulting state. The converse fails: commutativity, the sixth-power relations, and the displayed kernel allow several words to reach the same state. Recovering a final state recovers neither the ordering of its operations nor the number of complete turns. An observation of history must retain the word or an appropriately defined accumulated record. This distinction persists even when all state calculations are exact. A value and the process that produced it are therefore separate records: equality of two values is decided on the state, and equality of two processes on their words.

6.2 Orthogonal observation and the retained sheet

Choose the real coordinate order in which the orthogonal observation and the omitted coordinate are

$$x = \frac{1}{\sqrt{2}}(\cos j\theta, \sin j\theta, \cos k\theta), \qquad w = \frac{\sin k\theta}{\sqrt{2}}. \tag{6.2}$$

The first two observed coordinates recover j exactly on the finite image. The last gives $\cos k\theta$, and hence determines k up to $k \leftrightarrow -k$. For $k = 0, 3$ there is only one possibility; for $k = 1, 2, 4, 5$ there are two. Retain the sign of $\sin k\theta$ on these double fibers, using the canonical symbol 0 on the singleton fibers. The sheet lift (2.2) followed by the inverse of E then recovers the state.

For each fixed j, the two singleton fibers are $\{0\}$ and $\{3\}$, while the double fibers are $\{1, 5\}$ and $\{2, 4\}$. Orthogonal observation therefore has 24 distinct values: twelve singleton fibers and twelve double fibers. The sheet is a receipt exactly where the observation identifies two states; its necessity precedes any numerical error, since two members of a double fiber have identical observations.

The 24 states with $k \in \{1, 2, 4, 5\}$ form the stable subdomain

$$D_* = \{(j, k) \in D : k \in \{1, 2, 4, 5\}\}.$$

Here $|w| = \sqrt{3/8}$, so the continuous lift has a positive conditioning margin. The other twelve states lie on the silhouette $w = 0$. They remain distinct admitted states with unique exact observations. The loss of a uniform continuous inverse bound at the silhouette concerns perturbations in the continuous observation domain and leaves their finite codes intact; the finite observation alphabet has positive separation there (Lemma 6.2).

6.3 Paired circle observations and their antipodal fiber

The paired Hopf maps give a second observation. On $E(D)$ their third coordinates vanish, and their equatorial phases are

$$u = j + k \pmod 6, \qquad v = j - k \pmod 6.$$

The observation is represented by $(e^{iu\theta}, e^{iv\theta}) \in \mathbb{C}^2$. Its image consists exactly of the 18 pairs with $u \equiv v \pmod 2$. For the inverse, take the canonical representatives of such a pair and set

$$j_0 = \tfrac{1}{2}((u+v) \bmod 6) \in \{0,1,2\}, \qquad k_0 = (u - j_0) \bmod 6. \tag{6.3}$$

The parity condition makes the division by two an ordinary integer division. We have $j_0 + k_0 \equiv u$ and $j_0 - k_0 \equiv v$ modulo six, so this construction reaches every allowed observation. The kernel calculation from the preceding section shows that its two preimages are

$$(j_0, k_0), \qquad (j_0 + 3, k_0 + 3) \pmod 6.$$

Their geometric encodings are antipodal. A retained bit $b \in \{0,1\}$ selects the state $(j_0 + 3b, k_0 + 3b)$.

The bit has a different role from the parity of the orbit. The observation already gives $j + k \equiv u \pmod 2$, so it identifies the even or odd sector. Its remaining ambiguity is the antipodal pair inside that sector. Likewise, the antipodal bit belongs to the state observation; it is distinct from a choice of lift of a rotation through the spin covering. The distinctions follow from the domains of the maps, though each ambiguity is represented by two symbols.

Exact reader on D	Image size	Largest fiber	Retained recovery data
Full code $j + 6k$	36	1	None
Orthogonal $P \circ E$	24	2	Sheet on double fibers
Paired Hopf phases	18	2	Antipodal bit
Constant quotient	1	36	A representative code

Table 6.1: Observation fibers of the finite realization. The retained data complete each reader to an injective observation.

For the last row, take the quotient generated by the relation $n \le n'$ on the 36 integer codes. Totality identifies every pair in the generated equivalence relation, so the quotient has one element. Every numerical function descended to it is constant. The augmented output $(*, n)$, however, retains an integer code and has an exact inverse. The integer belongs to the larger output and is not reconstructible from the collapsed quotient.

6.4 Separation and finite decoding

Lemma 6.1 (Nearest-image decoding)**.** *Let $A \to Y$ be an injective finite encoding into a metric space, with minimum separation $s > 0$ between distinct image points. An observation at distance strictly less than $s/2$ from an image point has that point as its unique nearest image.*

Proof. If $d(y,a) < s/2$ and $a' \neq a$, then $d(y,a') \ge d(a,a') - d(y,a) > s/2$. □

Strict inequality is required: a midpoint can have two nearest images. Lemma 6.1 is the whole of the decoding theorem used: no property of the encoding beyond s enters, and no further result is invoked by name.

Lemma 6.2 (Separations of the finite realization)**.**

$$\min_{P(E(s))\neq P(E(s'))} \|P(E(s)) - P(E(s'))\| = s_P = \frac{1}{2\sqrt{2}},$$
$$\min_{\mathcal{H}(E(s))\neq \mathcal{H}(E(s'))} \|\mathcal{H}(E(s)) - \mathcal{H}(E(s'))\| = s_H = \sqrt{2},$$
$$\min_{s\neq s'} \|E(s) - E(s')\| = s_E = \frac{1}{\sqrt{2}},$$

the paired image represented by the phases $(e^{iu\theta}, e^{iv\theta})$, which on $E(D)$ are the complex components of $\mathcal{H}$. Nearest-image decoding is unique on open balls of radius $s_P/2$, $s_H/2$, and $s_E/2$ respectively.

Proof. If two observations have different first indices, their first two coordinates differ by at least $1/\sqrt{2}$, the chord length of one sixth-circle step after scaling. If their first indices agree, the distinct last coordinates are the four scaled values $1, 1/2, -1/2, -1$; their smallest spacing is $1/(2\sqrt{2})$. This gives s_P, valid on all 24 image points including the silhouette states. For the paired phase image, an allowed nonzero index difference has coordinates of the same parity. If both differences are odd, their squared chord lengths sum to at least two. If an even difference is nonzero, that coordinate contributes at least three. This gives s_H, attained by a simultaneous one-step change in both phases. For $E(D)$, a change in one clock by one step attains $1/\sqrt{2}$, and every nonzero coordinate change contributes at least $1/2$ to the squared distance of the scaled pair. The decoding radii follow from Lemma 6.1. □

After decoding, the receipts remain necessary. The sheet selects the admitted representative of an orthogonal double fiber. Switching the antipodal bit changes the recovered state by distance two and leaves the exact paired observation unchanged; no decrease of a continuous observation error resolves an exact identification. There are two recovery routes, with distinct constants: decode a finite observation and apply its receipt, or lift continuously within the valid domain and decode the finite state.

6.5 A worked state and a deterministic error bound

Consider $(j, k) = (1, 2)$. Start from the odd-sector seed $(1, 0)$, apply the negative S_- step with parameter $-\theta$, and then apply S_+ with parameter θ. Their index increments are $(-1, +1)$ and $(1, +1)$, so the resulting change is $(0, 2)$. The exact observation and omitted coordinate are

$$x = \left(\frac{1}{2\sqrt{2}}, \frac{\sqrt{3}}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}}\right), \qquad w = \frac{\sqrt{3}}{2\sqrt{2}} > 0.$$

The orthogonal receipt is + and the integer code is $13 = 1+6\cdot 2$. The paired phases are $(u, v) = (3, 5)$, for which (6.3) gives $j_0 = 1$, $k_0 = 2$ and receipt $b = 0$. The code is determined by $n = j + 6k$ and is identified with no other occurrence of the integer 13.

Theorem 6.3 (Exact recovery of the worked state)**.** *Let $s = (1, 2) \in D$. Suppose each computed round operation maps S^3 to itself with chord error at most 10^{-4} relative to its exact action, and observation quantization adds Euclidean error at most 10^{-3}. Then the perturbed observation satisfies $\|\widehat{x} - x\| \leq 0.0012 < \delta_*$, with δ_* as in Corollary 6.4, and the positive sheet lift L_+ followed by nearest-state decoding recovers $s = (1, 2)$ exactly.*

The error inputs are deterministic bounds, not measured standard deviations or outputs of a physical apparatus.

Proof. The exact operations are isometries. Applying (5.2) to the two-step word gives final geometric error at most $2 \cdot 10^{-4}$. Orthogonal projection is nonexpansive, so the resulting observation $\widehat{x}$ satisfies

$$\|\widehat{x} - x\| \leq \frac{1}{1000} + \frac{2}{10000} = \frac{3}{2500} = 0.0012. \tag{6.4}$$

Because $\|x\| = \sqrt{5/8}$ and $\sqrt{5/8} + 3/2500 < \sqrt{3}/2$, both observations lie in the closed ball of radius $\sqrt{3}/2$. The straight segment between them also lies there. Proposition 2.10 therefore applies with the conservative margin $a = 1/2$ and bounds the lifted-state error by $3/1250 = 0.0024$.

This is strictly below the state decoding radius $s_E/2 = 1/(2\sqrt{2})$, and $0.0012 < \delta_* \approx 0.0754$. Nearest-state decoding recovers $(1, 2)$. □

The conclusion uses four hypotheses: the operation bound, the observation bound, the sheet, and the admitted alphabet. An observation residual alone supplies none of the last two.

Corollary 6.4 (A neighborhood of exact finite recovery)**.** *For the state* $(1, 2)$*, retain the positive sheet and define*

$$\delta_* = \min\left\{\frac{\sqrt{3}}{2} - \sqrt{\frac{5}{8}}, \frac{1}{4\sqrt{2}}\right\} > 0.$$

Every observation y *with* $\|y - x\| < \delta_*$ *has a well-defined positive lift whose unique nearest admitted state is* $E(1, 2)$*.*

Proof. The first term in the minimum places y and x in the lift domain with margin $1/2$. Their lifted distance is therefore at most $2\,\|y - x\|$. The second term makes this distance strictly less than half the minimum state separation. Unique nearest-state decoding follows from the triangle inequality. □

The radius is fixed jointly by the distance to the lift boundary and the finite state spacing. A different sheet fails the hypothesis at zero projected error.

6.6 A calibrated period record

Attach to each canonical representative the cochain

$$h_{j,k} = (j, k, j + k)$$

on the torus complex (4.1). Its coordinates have units of dimensionless clock ticks, with one tick corresponding to angle θ. Since its third coordinate is the sum of the first two, it belongs to the harmonic space computed in Chapter 4. The period map $M(x, y, z) = (x, y)$ reads its two independent coordinates.

The map $D \to \mathbb{R}^3$ is a canonical phase record; crossing a clock wrap changes the printed representative. It is not a continuous real-valued angular coordinate on the whole torus. An accumulated winding record instead retains integer turns and depends on history. The present harmonic cochain records the admitted final state. An accumulated record, when one is kept, is the sum of the per-step index increments along the retained operation word; the final state alone does not determine that sum, by the kernel calculation of Section 6.1. Indeed its first two coordinates already determine that state, so this accompanying record constitutes additional retained information rather than an independent proof of an observation's injectivity.

For the worked state, introduce the raw record

$$\alpha = (1, 2, 3) + \frac{1}{1000}(1, 1, -1) = (1.001, 2.001, 2.999).$$

Writing $v = (1, 1, -1)$ gives $\rho = \langle \alpha, v \rangle = 3/1000$. The face residual is $(\rho, -\rho)$ and has norm $3\sqrt{2}/1000$. Orthogonal harmonic correction subtracts $\rho v/3$, returning $h = (1, 2, 3)$ exactly. The removed component has norm $\sqrt{3}/1000$, equal to the spectral-gap bound of Proposition 5.1. Equality holds because the complex has a single nonzero spectral direction, with eigenvalue six.

Read the two periods after this correction, and suppose their data are $m = (1, 2) + \xi$ with $\|\xi\| \leq 10^{-4}$. Reconstruction takes the explicit form

$$\widehat{h} = (m_1, m_2, m_1 + m_2).$$

For $\xi = (\xi_1, \xi_2)$,

$$\left\|\widehat{h} - h\right\|^2 = \xi_1^2 + \xi_2^2 + (\xi_1 + \xi_2)^2 \leq 3 \|\xi\|^2,$$

so $\left\|\widehat{h} - h\right\| \leq \sqrt{3}/10000$. The factor $\sqrt{3}$ agrees with the reciprocal harmonic response derived earlier. Reading the periods of the uncorrected α retains the bias $(1/1000, 1/1000)$; the order of correction and measurement is part of the reconstruction.

6.7 The combined reconstruction

Take the common observable to be

$$q(j, k) = (P(E(j, k)), h_{j,k}) \in \mathbb{R}^3 \oplus \mathbb{R}^3,$$

using the declared dimensionless coordinate norms. The product observation includes geometric data and the accompanying period record. The latter is injective on D, so the combined observable is injective as well. Agreement between its representations is a correspondence assertion for these specified maps; it is not evidence that the two records were acquired independently. In the same way, that a calculation has been carried out and has terminated is a statement about the calculation; what it establishes is the proposition it computes, which is compared with the claim on its own terms and not through the name of the record that reports it.

For the constructed example, (6.4) and the period estimate give, in the Euclidean product norm,

$$\left\|(\widehat{x}, \widehat{h}) - q(1, 2)\right\| \leq \frac{3}{2500} + \frac{\sqrt{3}}{10000} < 0.001374.$$

The sum is a conservative upper bound on the product norm and requires no probabilistic independence assumption. A common-observable allowance of 0.0014 and a lifted-state allowance of 0.0025 both contain the calculated errors. The same hypotheses also yield exact nearest-state decoding, as proved above.

A pair of cyclic indices determines a polytope vertex and a round state; the generators act compatibly on both; each observation has an explicit fiber and receipt; the harmonic record has a known local defect and a stable two-period reconstruction. Every recovery statement is an inverse, a separation margin, or an operator bound.

Chapter 7

Enclosure and recoverability

7.1 The completed correspondence

The whole is explicit: $F = \partial(P_6 \times P_6)$, with a stated cell structure and a homeomorphism to S^3. Its round motions are explicit: left and right phase multiplication. Its observations are explicit: orthogonal projection, individual and paired Hopf maps, and period readers on a selected complex. Their fibers specify which information must accompany an observation if a representative is to be recovered.

Theorem 7.1 (Finite representation correspondence)**.** *Let $D = (\mathbb{Z}/6\mathbb{Z})^2$, with canonical representatives $0 \le j, k < 6$, let $G = \langle S_+, S_- \rangle$ act on D, and let E_F, E be the encodings of* (6.1)*. Each of the following is an injective description $A : D \to A(D)$, with an explicit inverse on its image:*

1. *the integer code $j + 6k$;*
2. *the polytope vertex $E_F(j, k)$ and the round state $E(j, k)$;*
3. *the orthogonal observation with its prescribed sheet receipt;*
4. *the paired Hopf observation with its antipodal receipt;*
5. *the harmonic phase record $h_{j,k} = (j, k, j + k)$.*

With G acting on $A(D)$ by $T \mapsto ATA^{-1}$, the five descriptions are pairwise isomorphic finite G-sets (discrete dynamical systems), the isomorphisms being $A' \circ A^{-1}$. On $E(D)$ the transported actions of $S_\pm$ are the restrictions of the round actions $C_\pm(\pi/3)$. The same holds for every finite translation of D.

Proof. Division with remainder inverts the integer code. The two sixth roots of unity invert the geometric encoding, and Ψ^{-1} returns the polytope vertex. Orthogonal recovery uses the two-sheet formula, with the canonical singleton convention at the silhouette. Equation (6.3) and the retained bit invert the paired observation. Reading the first two coordinates inverts the harmonic record. All these formulas are established in Chapter 6. Multiplication by $e^{i\pi/3}$ in the specified complex factors proves the intertwining identity. For descriptions A, A', $(A'A^{-1})(ATA^{-1}) = (A'TA'^{-1})(A'A^{-1})$, so $A'A^{-1}$ is G-equivariant. □

Remark 7.2. The theorem concerns augmented observations: items three and four retain different receipts because their unaugmented maps identify different states. It concerns final configurations; many operation words reach one state, and a history reader requires further retained information. The harmonic record contains the two phase indices, so its coexistence with the geometric observation is a comparison of descriptions; statistical independence of records is a separate acquisition premise.

7.2 Exact structure and quantitative response

Two kinds of control enter. The first is exact structure: the specified sphere homeomorphism, the group relations, the incidence identity $d_1 d_0 = 0$, and injectivity of the retained data. The second is quantitative response: the sheet margin, the finite separation distances, the spectral gap, and the smallest harmonic measurement response. Exact structure makes a recovery question well-defined. A positive response bound controls its behavior under perturbations.

A correct sheet with a small margin can amplify a small geometric discrepancy substantially. A wrong sheet can produce zero projected discrepancy and still recover a different state. Likewise, a cochain with zero local residual may have an arbitrary harmonic component until its periods are measured. The spectral gap controls the component orthogonal to that kernel; the harmonic response controls the remaining component. The reconstruction estimates of Chapter 5 make this division explicit.

For the worked state, the declared operation and observation errors produce an observation bound of 0.0012 and a same-sheet lift bound of 0.0024. The period perturbation contributes at most $\sqrt{3}/10000$. Their stated common-observable bound is less than 0.001374, and the lifted state lies strictly inside the unique-decoding radius. Continuous estimates yield exact recovery of a finite state, from the admitted alphabet and the margins together with the small residuals (Theorem 6.3).

Consistency of descriptions has a precise meaning in this model. Their exact diagrams commute, and their perturbed readings lie within bounds derived from the same specified maps. Such agreement is a result about a mathematical representation and its declared inputs. A physical application adds the task of establishing that those inputs and observation maps describe the apparatus at its calibrated resolution. The mathematical comparison can then be used as part of that application without changing the domain of its theorem.

7.3 Extensions of the construction

The product of two hexagons was chosen to make the twelve prism facets and the six-position clocks coincide exactly. Products of other polygons offer a natural extension. Their boundaries remain three-spheres after the corresponding factorwise disk construction. The arithmetic of the selected finite steps depends on the clock size: the same phase matrix is invertible modulo an odd size, whereas an even size retains an index-two obstruction. The twofold ambiguity of the continuous torus map still requires its own lift. Discrete sampling and continuous observation are compared for the chosen alphabet; neither is inferred from the other.

A limit of refined alphabets requires its own response analysis. With N equally spaced positions in each factor and the same round normalization, the least state separation is

$$s_N = \sqrt{2}\sin(\pi/N).$$

A one-step change in either factor attains this distance, and every distinct pair has at least that separation. Unique nearest-state decoding is therefore ensured by a state error strictly below $\sin(\pi/N)/\sqrt{2}$. The radius tends to zero as the alphabet is refined. A continuous inverse estimate can remain valid on a fixed sheet margin while the noise allowance for recovering an *exact discrete label* shrinks. The target of reconstruction must be specified when taking such a limit. A repeated recovery at a refined alphabet is therefore a different experiment from the six-position one, with its own allowance; restating the six-position result does not perform it.

For nested admitted domains, extending the exact correspondence also requires compatible encodings and readers: encoding a state before inclusion must agree with including its encoding, and

the observation diagrams must commute on the shared states. Uniform stability asks, in addition, which inverse constants remain bounded across the family. A result for one finite domain does not supply the limiting answer.

A second extension concerns richer measured complexes. Refinement introduces more cochain coordinates and requires compatible maps between complexes, controlled inner products, and estimates uniform in the chosen family. The transport argument shows what a pure change of coordinates preserves; a convergence argument for refinement would supply additional analytic control [1].

A third extension puts a field on the edges. The argument card (its Section 9) lays squares on a torus of any size and gives each edge a link, a quaternion with whole entries. Turning every vertex by a link of length one conjugates each square's holonomy at its corner: the whole part of the holonomy is retained exactly, the three turned parts are retained up to a rotation, and an action that reads only the turned parts is unchanged. Flat configurations are stationary on every edge. *Recorded* from the argument card, where each statement is checked by the middle card, at every size, and the same holds on a lattice of four directions with a square for every pair. Each torus is taken on its own: the compatible encodings, readers and estimates across the family, which a limit needs, are those of the paragraphs above, and the argument card does not supply them.

A physical realization would specify a measured quantity for each relevant coordinate, the apparatus that reports it, and the conversion between geometric or cochain discrepancy and measurement uncertainty [11, 10]. The enclosure would then participate in a measurement model with declared scales and a testable response. The present construction supplies the geometric and algebraic maps on which such a model can be built.

Enclosure fixes the state space; operations specify how its states change; observation fibers identify what is lost; retained lifting data make an inverse possible; and calibrated response bounds determine how reliably that inverse can be used. On the finite sector developed here, every passage is explicit.

7.4 The shell as a slice

The constructions above concern a configuration space with a positive definite metric. This section records one proposed reading of that space as a spatial slice of a spacetime. The facts it rests on are standard and cited; the identifications built on them are calibrated *proposed* in the claim ledger. No theorem of the preceding chapters depends on them, and none establishes them.

De Sitter space of radius ℓ is the one-sheeted hyperboloid $-X_0^2 + X_1^2 + \cdots + X_4^2 = \ell^2$ in five-dimensional Minkowski space. In its global slicing the metric is

$$\mathrm{d}s^2 = -\,\mathrm{d}t^2 + \ell^2 \cosh^2(t/\ell)\ \mathrm{d}\Omega_3^2, \tag{7.1}$$

so each slice of constant t is a round three-sphere of radius $\ell \cosh(t/\ell)$. The spacetime is maximally symmetric. An observer at rest sees a time-independent static patch bounded by a cosmological horizon, a two-sphere at static radial coordinate $r = \ell$ (areal radius ℓ), whose thermodynamic properties were computed by Gibbons and Hawking [5]. Identifying antipodal points of the hyperboloid gives the elliptic interpretation of de Sitter space, in which the static patches of two antipodal observers become one [14].

The normalized frinkahedron $\Psi(F)$ is a round three-sphere. The proposed reading consists of the following identifications.

1. The sphere $\Psi(F)$, scaled by $\ell \cosh(t/\ell)$, is the global slice of (7.1) at time t. An apparent expansion is then the growth of the slicing index; the hyperboloid itself does not change.

2. The orthogonal observation of Section 2.4 is the view from the static patch of an observer at the pole $w = 1$: the open sheet $w > 0$ is that patch, the silhouette $S^2 = \{w = 0\}$ is its horizon, and the opposite sheet $w < 0$ is the patch of the antipodal observer.
3. Retaining the sheet sign keeps the two patches distinct, as in de Sitter space. The elliptic quotient identifies z with $-z$; it carries the sheet $w > 0$ onto the sheet $w < 0$ and so has a single patch. On this slice, away from the two core circles, the elliptic identification is exactly the fiber of the paired Hopf observation, Proposition 3.3. Thus the retained antipodal bit is the datum that separates de Sitter space from its elliptic quotient, and the sheet sign is the datum that separates the two patches of the former.
4. The counter of the record in Chapter 8, which increases by one per reply, is global time on the stack of slices. It increases because the record is extended by one entry per reply, and for no other reason.

Each identification is testable against the cited facts. None is used in this volume. The companion tolerances volume carries the anti-de Sitter and de Sitter continuation reading of the two records, the review and the typeset record. The de Sitter reading needs a positive cosmological constant, and the companion volume supplies it without leaving its degree count. In its treatment of first-order vacuum gravity, under *The cosmological term as a residue*, the proposition *The cosmological constant as an integration constant* restricts the metric density to the unimodular sector, keeps the cubic action unchanged, and obtains $R_{\mu\nu} = \Lambda g_{\mu\nu}$, with Λ a constant left by the contracted Bianchi identity and carried unchanged by the solution [15, 8, 16]. De Sitter space with $\Lambda > 0$ is then a solution of that gravity model, and (7.1) is its global slicing; the identifications listed above remain proposed. Here the configuration coordinates remain configuration coordinates until a calibration connects them to an observable, as stated at the opening of Chapter 2.

Chapter 8

The record and the press check

8.1 The record

Definition 8.1 (The table)**.** The *table* is the triple of cards: the geometric card (this volume, retention), the mechanized construction (the middle card, execution), and the numerical card (the companion tolerances volume, what may be claimed from a reading).

The middle card ends with two acts. In the first, the review, a reviewer makes thirty-six demands, numbered as pages from thirty-six down to one. At each page the reviewer pairs a claim found by search against the same claim handed over by name and extracts the fact each carries. In the second, the typeset record, a publication is built upward from page one to page thirty-six; each page carries three things unchanged in kind, and one thing is decided at the top. Each side volume is required to perform both acts in its own mathematics: to answer the thirty-six demands directly, and to carry the record's three things and read its verdict.

This volume fixes what must be retained: the fourth coordinate, the sheet sign, the Clifford seam, the certified faces. The numerical card fixes what may be claimed from a reading: units, scales, stability, an error budget, a reporting cell. The middle card and this chapter check that the record carries both. Closure of the actions and compilation of the construction are necessary for the agreement the cards report and not sufficient for its meaning. The register below audits whether the record retained what its own theorems require.

This chapter performs both acts. It adds no theorem; every statement cites a result proved, cited, or proposed earlier in this volume, with the comparison the review asks for. Four definitions fix its structure.

1. *The register* is the invariant tuple

$$\mathbf{Reg} := ((1,2),\ 13,\ +,\ 0) \in D \times \mathbb{N} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z},$$

   the worked state of Section 6.5, its integer code $13 = 1 + 6 \cdot 2$, its sheet sign, and its antipodal bit. It is carried unchanged through every entry and reported at the end as the *value*; every observation of Chapter 6 recovers it.

2. *The counter* is the monotone index $k \in \{1, \dots, 36\}$ of the entries in typeset order, increased by exactly one per entry and never skipped. Its top value, 36, is reported as the *velocity*. Entry n answers review page n, so the record ascends the pages that the review descended. The count of entries is the count of demands; the count of states, also thirty-six, is $6 \cdot 6$, and the two are not identified. The counter also names the assumption it does not make. The field with one element has one element, every map is the identity on it, and counting it costs nothing:

that field is, as this record reads it, the assumption that counting its elements takes zero steps instead of one. Writing an entry costs one step, the record grows by exactly one entry per step, and the counter is the tally of those steps, so its top value is the cost of writing this record and not of counting it. Appending to a deterministic record can leave its Shannon uncertainty unchanged while its length, and the steps taken to write it, increase by one per entry.

3. *The trace* is the sequence $(c_k, s_k)_{k=1}^{36}$ of Section 8.2, accumulated in order: c_k, the construction this volume proves or derives, with its citation, and s_k, the reading handed over by name on the same point. Either s_k quotes the middle card's patter, with that source named, or it is a *comparison statement constructed for the review*, which states in the vocabulary of this volume the reading that the obligation puts in question. Each entry writes $c_k \leftrightarrow s_k$ and does not decide it. No later entry re-reads or re-decides an earlier one.
4. *The verdict* is read once, at the top, in Section 8.4: two readings of the register that this volume proves, and the triple (slipped, value, velocity) computed from them.

Each citation carries a calibration label, and the label names the reference against which the citation is read: *proved here*, a proof in this volume; *standard*, a cited source; *measured*, the inspected declarations of the middle construction, an extraction from source and not an experiment; and *proposed interpretation*, the knower's identification. The entries use the first, second, and fourth. Every number of this volume is derived in the text, so no entry is calibrated measured. The lifts of Section 8.3, which read the middle card's own counts, use all four.

8.2 The thirty-six replies

Entry k answers review page k. The obligation is the reviewer's demand, restated without reference to the construction in which it was made. Review pages 24, 30, and 31 extract no fact by construction; their entries are presence entries, which locate rhetoric, universality, or deduction in this volume and write $\mathbf{id} \leftrightarrow \mathbf{id}$. A reading s_k marked *constructed for the review* is a comparison statement constructed for the review; every other s_k names its source.

Entry 1. *Obligation.* Exhibit the single distinguishable mark on which the whole publication rests and compare it with the mark the record started from; the descent stops here and nothing beneath it is written.
Citation. The positive minimum separations of Section 6.4: $s_P = 1/(2\sqrt{2})$ for the orthogonal image, $s_H = \sqrt{2}$ for the paired image, and $1/\sqrt{2}$ for the state image, each with strict $s/2$ decoding. Section 6.2 shows the necessity of a mark before any numerical error: two members of a double fiber have identical observations. The closing paragraphs of Section 5.7 record that a norm inequality supplies no confidence level and that these failures precede any tolerance.
Calibration. Proved here.
Constructed c_1. The mark is a positive separation $s > 0$ of a finite image. Distinguishability of two admitted states is the inequality $s > 0$ and nothing else; the record of this volume starts from the thirty-six admitted pairs and the separations they are proved to have.
Handed over s_1. A mark is distinguishable because it has been seen in a drawing, and the record starts from that drawing (constructed for the review).
Written, not decided. $c_1 \leftrightarrow s_1$.

Entry 2. *Obligation.* State the local rule under which the mark was admitted into the record and compare the carried fact on file with the symbol of the carrier so admitted.
Citation. The opening of Section 6.1: $D = (\mathbb{Z}/6\mathbb{Z})^2$ with canonical representatives $0 \le j, k < 6$,

stated as the rule of admission for the finite record, and the encodings (6.1). Section 1.1 fixes the admitted clock sector as the meaning of "the whole" for the finite realization, and the introduction of Chapter 6 names the sector a finite sector of the continuous enclosure.
Calibration. Proved here; the rule is a definition and the injectivity of the carrier is proved.
Constructed c_2. A state is admitted as a pair of residues with canonical representatives. The carrier of the admitted pair is the encoding E, which is injective, so the carried fact on file is the pair itself.
Handed over s_2. A state is admitted once an admitted set and an encoding are named, without a rule of admission, and the carrier symbol of a mark admitted by a counting process is the mark itself (constructed for the review).
Written, not decided. $c_2 \leftrightarrow s_2$.

Entry 3. *Obligation.* Counting must count a concrete indexed carrier, not merely name a class; the fact on file is compared with the symbol of the carrier being counted.
Citation. Proposition 3.4: the group generated by S_+, S_- has eighteen elements and two orbits, distinguished by $j + k$ modulo two. The sector paragraph of Section 6.1 computes the kernel $\{(0,0),(3,3)\}$ and the image. Proposition 2.2 counts the cells of F.
Calibration. Proved here.
Constructed c_3. The count eighteen counts the image of $(a,b) \mapsto (a+b, a-b)$ on $(\mathbb{Z}/6\mathbb{Z})^2$, a group with a kernel of order two; the count thirty-six counts ordered pairs of hexagon vertices; twelve counts prism facets. Each number is attached to the indexed set it counts.
Handed over s_3. A class named is a carrier counted, and an Euler-characteristic check that counts cells without their incidences counts the carrier (constructed for the review). The middle card's patter supplies the named classes: it speaks of "twelve fenceposts" and names two functions "once around and twice around".
Written, not decided. $c_3 \leftrightarrow s_3$.

Entry 4. *Obligation.* Encoding must preserve a read path back to the original mark: follow the limit process through its index to the carrier symbol it depends on and compare with the fact on file.
Citation. Section 6.1: both encodings are injective, a sixth root of unity determines its exponent modulo six, and division with remainder inverts the integer code. The basic recovery equation (1.2). Theorem 7.1 lists an explicit inverse on the image of each description.
Calibration. Proved here.
Constructed c_4. The read path from $E(j,k)$ back to (j,k) is the reading of two exponents; from n back to (j,k) it is $(n \bmod 6, \lfloor n/6 \rfloor)$. No limit process is needed on the finite image.
Handed over s_4. Recovery holds on the chosen recovery domain without an exhibited inverse, and an encoding may reach its mark through a limit process indexed down to a carrier symbol (constructed for the review).
Written, not decided. $c_4 \leftrightarrow s_4$.

Entry 5. *Obligation.* Residue must be a limit record, indexed down to the counted carrier, not a leftover narrative; the fact on file is compared with the symbol of the carrier at the end of that index.
Citation. Proposition 5.1 and the paragraph following it: the transverse component is bounded by $\lambda^{-1/2}$ times the residual norm, and no decrease of a residual tolerance removes the harmonic kernel. Section 5.3 computes $e = (\rho/3)(1,1,-1)$ with $\|e\| = |\rho|/\sqrt{3}$. Section 6.6 exhibits $\rho = 3/1000$, exactly at the gap bound.
Calibration. Proved here.

Constructed c_5. The residue is the transverse component e of a cochain, an exact vector computed on the complex, with the bound it satisfies and the harmonic component it cannot determine.
Handed over s_5. The residue is what a counting process leaves over, projected to a carrier symbol, and local residuals are bounded without the operator whose spectrum bounds them (constructed for the review).
Written, not decided. $c_5 \leftrightarrow s_5$.

Entry 6. *Obligation.* Record where the binary split is written after execution by reducing both bits, initial condition and signal response, to the still-readable carrier fact, and restore to the argument the exhibit the submission constructed and then dropped.
Citation. Proposition 3.3: the binary receipt ϵ after a section convention. Section 6.3: the bit b selects $(j_0 + 3b, k_0 + 3b)$ through (6.3). Section 2.4: the sheet receipt σ of (2.2). Table 6.1 lists both bits beside the observations they complete.
Calibration. Proved here.
Constructed c_6. Two bits, written beside two different observations: σ, the sign of w, and b, the antipodal choice. Each is retained with its observation and consumed by the displayed inverse formula; neither is dropped from any later argument of this volume.
Handed over s_6. A binary exhibit, once constructed, may be omitted from every later argument, and retaining a sheet sign and retaining a binary lift label are two instructions without the inverses they feed (constructed for the review).
Written, not decided. $c_6 \leftrightarrow s_6$.

Entry 7. *Obligation.* Repetition must re-run the experiment, an expectation with a signal response, not restate the sentence; the expectation's fact is compared with the trial fact on file.
Citation. Section 7.3: with N positions per clock the least separation is $s_N = \sqrt{2}\sin(\pi/N)$, the decoding radius shrinks, and a recovery at a refined alphabet is a different experiment with its own allowance. Section 4.3: a bounded discrepancy is not an equivalence relation. Section 5.7: deterministic bounds carry no coverage interpretation.
Calibration. Proved here for s_N and the decoding radius; standard, cited, for the distinction between a bound and a coverage statement.
Constructed c_7. The expectation is the decoding radius $\sin(\pi/N)/\sqrt{2}$; the signal response is whether the lifted error lies below it. At $N = 6$ the worked state's lifted error $3/1250$ does, and the trial recovers $(1, 2)$.
Handed over s_7. Repetition is the sentence that the result shows up every time, and discrepancies are compared against the sum of both representations' certified error budgets without a re-run (constructed for the review).
Written, not decided. $c_7 \leftrightarrow s_7$.

Entry 8. *Obligation.* Show what the number counts: trace both numeric carriers down to the fact each was packaged with, a hypothesis or a tally.
Citation. Proposition 2.2 and Table 2.1; the example on dimension and incidence in Section 2.1, where the count twelve acquires its meaning through dimension and incidence; Table 6.1, with image sizes $36, 24, 18, 1$.
Calibration. Proved here.
Constructed c_8. Thirty-six counts ordered pairs of hexagon vertices; twelve counts three-dimensional prisms; twenty-four and eighteen count observation fibers. Each number is a tally of a listed set.
Handed over s_8. The twelve fenceposts and the thirteen read from a tape, each a number packaged with a hypothesis, are tallies (constructed for the review).

Written, not decided. $c_8 \leftrightarrow s_8$.

Entry 9. *Obligation.* A representation must have a carrier state, the computation state at the end of the representation chain, not merely a notation; that state's fact is compared with the fact on file.
Citation. Section 3.4: the representation is the data through which an operation acquires a geometric meaning. Section 6.1: three codomains for the same admitted states, with explicit maps between their images. Section 4.6: what a change of representation carries.
Calibration. Proved here.
Constructed c_9. The carrier of the integer code n is the state $(n \bmod 6, \lfloor n/6 \rfloor)$; of E, a point of the Clifford torus; of $h_{j,k}$, a harmonic cochain. Each is a state with an inverse, not a notation.
Handed over s_9. A representation ends in a computation state named by a file, and a Clifford formulation is a representation before any representation assigns its symbols to operators (constructed for the review).
Written, not decided. $c_9 \leftrightarrow s_9$.

Entry 10. *Obligation.* The physical claim must exhibit its own threshold fact, halting or non-halting, rather than inherit authority from higher levels; the two thresholds' facts are compared.
Citation. Section 6.4: the thresholds $s_P/2 = 1/(4\sqrt{2})$, $s_H/2 = 1/\sqrt{2}$, and $1/(2\sqrt{2})$ for the state image. The closing paragraphs of Section 7.3 and the opening of Chapter 2: a physical realization requires a calibration that this volume does not supply.
Calibration. Proved here for the thresholds. No physical claim is made; the physical readings of this volume are calibrated proposed interpretation.
Constructed c_{10}. Three decoding thresholds, each an inequality decided on the data. The worked lift error 0.0024 lies below $1/(2\sqrt{2})$; that inequality is the threshold fact, and it is the volume's own.
Handed over s_{10}. A configuration space read as a slice of a spacetime inherits its physical authority from the geometry (constructed for the review). This volume lists that reading in Section 7.4 as proposed and derives no result from it.
Written, not decided. $c_{10} \leftrightarrow s_{10}$.

Entry 11. *Obligation.* Comparability requires a common physical value process, a base and a step, relating the two quantities; naming them alike is not comparing them.
Citation. Proposition 5.5 and (5.3): comparison through one observable q on one admitted domain, with a four-term budget. Section 6.5: thirteen follows from $1 + 6 \cdot 2$, and no correspondence with a separately defined thirteen-count tape follows from the coincidence. Section 1.4.
Calibration. Proved here.
Constructed c_{11}. The base is the observable q; the step is a reader's Lipschitz constant applied to a state error; the comparison is the inequality (5.3) with its four terms.
Handed over s_{11}. A tape that reads thirteen and the worked state's code thirteen, placed side by side, are compared by their equal names (constructed for the review).
Written, not decided. $c_{11} \leftrightarrow s_{11}$.

Entry 12. *Obligation.* An observation must be a projection with a stated dimension, what was projected onto what, not a report of seeing; its fact is compared with the fact on file.
Citation. Proposition 2.10 and (2.2) in Section 2.4. Section 2.3 separates three uses of dimension. Equation (6.2) names the omitted coordinate w for the finite realization.
Calibration. Proved here.
Constructed c_{12}. $P : S^3 \to \overline{B}^3$ deletes the fourth coordinate: a three-manifold in $\mathbb{R}^4$ is projected onto a filled three-ball, with fibers of size two off the silhouette and size one on it.
Handed over s_{12}. A picture that encloses a volume is an observation, with the omitted coordinate

unnamed (constructed for the review). The volume enclosed is the image's, not the object's, by Section 2.3.
Written, not decided. $c_{12} \leftrightarrow s_{12}$.

Entry 13. *Obligation.* State where instantaneous presence accumulates into a record, the increment tree or the integration over an instant, and compare the accumulated fact with the fact on file.
Citation. Section 6.6: the harmonic cochain records the admitted final state; an accumulated record is the sum of per-step index increments along the retained word, and the final state alone does not determine that sum. Section 4.4: the periods on a, b, c. Section 2.5.1: the volume integrals.
Calibration. Proved here.
Constructed c_{13}. Accumulation is the sum of index increments along the retained word; for the worked state the increments $(-1, +1)$ and $(1, +1)$ sum to $(0, 2)$. The present record $h_{1,2} = (1, 2, 3)$ is not an accumulation.
Handed over s_{13}. Presence accumulates through a tree of increments or an integral over an instant, and an accumulated state error $\sum_j \epsilon_j$ is that accumulation (constructed for the review). The sum accumulates error, not presence.
Written, not decided. $c_{13} \leftrightarrow s_{13}$.

Entry 14. *Obligation.* Measurability must be anchored to an observable clock, a gauge process with a stated reference, whose fact is compared with the fact on file, rather than asserted.
Citation. Equation (3.4) and Figure 3.2: the vertical maps specify which continuous states the clock labels represent. Section 6.1: $S_+ = T_1 T_2$ and $S_- = T_1 T_2^{-1}$ agree with the round actions at parameter θ as an intertwining identity.
Calibration. Proved here.
Constructed c_{14}. The clock is $\mathbb{Z}/N\mathbb{Z}$ acting by $e_z(k)$; the reference is the state z; the gauge is the commuting diagram. For the finite realization the clock is $(\mathbb{Z}/6\mathbb{Z})^2$ with the stated steps.
Handed over s_{14}. Assigning the permutation clock to one family and the continuous description to the other exhibits a clock (constructed for the review). The assignment divides roles; it names a timestamp.
Written, not decided. $c_{14} \leftrightarrow s_{14}$.

Entry 15. *Obligation.* Even the auxiliary or comic concept class must account for its evidence: whichever concept it invokes is reduced to the fact it reports and compared with the fact on file.
Citation. The figurative names of this volume reduce to defined objects. Definition 2.1: the frinkahedron is $\partial(P_6 \times P_6)$. Definitions 1.1 and 4.1: a receipt is a map R injective on observation fibers. The carried number of this record, Section 8.1, is the integer code 13 of the worked state. The one-point quotient of Section 4.2, the constant-quotient row of Table 6.1, and the closing paragraph of Section 6.3 show that a collapsed name reports nothing the retained integer did not.
Calibration. Proved here; the reductions are definitions.
Constructed c_{15}. Each name reports exactly the object it abbreviates: a polyhedral boundary, a retained lifting datum, an integer. No evidence is carried by the name.
Handed over s_{15}. A concept named for a color, a bang, or a superposition carries evidence by its name, and so does the frinkahedron as the name of a drawing (constructed for the review). The middle card's patter counts "all 12 faces of our Frinkahedron".
Written, not decided. $c_{15} \leftrightarrow s_{15}$.

Entry 16. *Obligation.* Identify which fact at the source survives transmission through the channel: the physical datum and the zero-like datum carry a fact; the unit-like branch carries none.
Citation. Proposition 4.4: a receipt alphabet needs at least m symbols on an m-element fiber,

$\lceil \log_2 m \rceil$ bits. Section 4.3: conditional entropies 2/3 bit, one bit, and $\log_2 36$ bits for the three readers, under the uniform distribution on D.
Calibration. Proved here.
Constructed c_{16}. Through the orthogonal channel, twenty-four of thirty-six distinctions survive; the receipt that restores the rest carries 2/3 bit on average under the uniform distribution, while its worst-case alphabet, fixed by the double fibers, has two symbols, one bit. Through the paired channel, eighteen distinctions survive, with one bit on average and in the worst case. Through the constant channel, none survives: $\log_2 36$ bits on average, and an alphabet of thirty-six symbols, six bits, in the worst case.
Handed over s_{16}. A channel that emits zero and one from a translation layer transmits its fact, and a Hopf readout forgets a whole phase circle, with no count of what survives (constructed for the review).
Written, not decided. $c_{16} \leftrightarrow s_{16}$.

Entry 17. *Obligation.* Execution must leave a record of the elementary operations performed; the fact on file is compared with the fact at the head of that record, and a gesture without a record is not execution.
Citation. Section 3.5: interpreting a reordered tape requires the permutation or its inverse. Section 5.5: a lost inverse permutation changes the exact word being compared. Section 6.1: a seed and a retained operation word determine the state.
Calibration. Proved here.
Constructed c_{17}. The record is the word $S_+^a S_-^b$ with its order. For the worked state it is the step S_- with parameter $-\theta$ followed by S_+, from the seed $(1, 0)$; its head is the last step applied.
Handed over s_{17}. A performed gesture whose elementary operations are recorded by name is an execution, and words retain their order without a word being written (constructed for the review).
Written, not decided. $c_{17} \leftrightarrow s_{17}$.

Entry 18. *Obligation.* Separate a value from the process that produced it and compare the facts carried by each, whatever mode, described, compiled, or executed, the process was recorded in.
Citation. Section 3.5: $E_z \Pi_j = C_+(2\pi j/N) E_z$ with a kernel of dimension $N - 2$, so agreement of sampled operations is not an isomorphism of ambient spaces. Section 6.1: a value and the process that produced it are separate records, equality of values decided on the state and equality of processes on words.
Calibration. Proved here.
Constructed c_{18}. The value is the state $(1, 2)$; the process is the two-step word. Equality of two values is decided in D; equality of two processes is decided on their words modulo the kernel $\{(0, 0), (3, 3)\}$ and the sixth-power relations.
Handed over s_{18}. A constant described, compiled, or executed is one record in three modes, and agreement of the sampled operations is an isomorphism of descriptions (constructed for the review).
Written, not decided. $c_{18} \leftrightarrow s_{18}$.

Entry 19. *Obligation.* Show the step at which magnitudes are actually added and compare its fact with the fact on file; the additive identity is not a sufficient claim.
Citation. Equation (5.2) and its remark in Section 5.5: isometries reduce the bound to addition of step errors. Section 6.7: the sum is a conservative upper bound on the product norm.
Calibration. Proved here.
Constructed c_{19}. The additions are $1/1000 + 2/10000 = 3/2500$ in (6.4) and $3/2500 + \sqrt{3}/10000 < 0.001374$ in Section 6.7.

Handed over s_{19}. The sum $\sum_j \epsilon_j$, named without being performed on an input, is the step at which magnitudes are added (constructed for the review).
Written, not decided. $c_{19} \leftrightarrow s_{19}$.

Entry 20. *Obligation.* A scale must be exhibited as the product that produces it, not merely by naming the unit; the multiplicative identity alone contributes nothing.
Citation. Section 2.5: the chart's magnification relative to round length is $(1 + \|y\|^2)/2 = 1/(1 - w)$, computed from the line element. Section 2.5.1: the volume forms and $\mathrm{Vol}(S^3) = 2\pi^2$. Section 5.2: rescaling an output component is a calibration choice.
Calibration. Proved here.
Constructed c_{20}. The scale factor is the product of the conformal factor with round length; the volume $2\pi^2$ is an integral evaluated by substitution.
Handed over s_{20}. A scale is exhibited by naming its unit, and an apparent enlargement read from a drawing is a change of size (constructed for the review). Section 2.5 attributes the enlargement to the chart.
Written, not decided. $c_{20} \leftrightarrow s_{20}$.

Entry 21. *Obligation.* Before any comparison, exhibit the basis, or the null space, in which the datum is expressed; an origin without a basis contributes no fact.
Citation. Section 4.5 and (4.2): $\mathcal{H}^1 = \{(A, B, A + B)\}$, with $v = (1, 1, -1)$ spanning the complement. Theorem 5.3: injectivity exactly when M is injective on $\mathcal{H}^1$. Section 5.2: one period cannot be injective on a two-dimensional harmonic space.
Calibration. Proved here.
Constructed c_{21}. The basis is $(1, 0, 1), (0, 1, 1)$ of $\mathcal{H}^1$; the null space of d_1 is $\mathcal{H}^1$ because $d_0 = 0$; the period map is injective there with response $\gamma = 1/\sqrt{3}$.
Handed over s_{21}. At least $\dim \mathcal{H}^1$ independent scalar period directions suffice without a basis, and a load that lands in a solver's span with the origin branch refused is expressed in a basis (constructed for the review).
Written, not decided. $c_{21} \leftrightarrow s_{21}$.

Entry 22. *Obligation.* State whether finiteness is imposed exactly, as a multiplier or boundary condition satisfying the inf-sup condition, or only penalized by a relaxation parameter; record the finite expansion at its fact-bearing terms, the bare monomial contributing nothing.
Citation. Section 4.4: closedness $x + y - z = 0$ is a condition a cochain may fail; it is enforced exactly by orthogonal projection onto $\ker d_1$, not through a penalty, and the harmonic representative is selected by the same exact projection. Section 4.5: a small cochain coefficient is information about a cochain, not an attachment of a new cell. Section 5.4: the shared-kernel hypothesis is essential.
Calibration. Proved here. The inf-sup condition itself does not arise in this volume; the companion volume answers the multiplier question for its models.
Constructed c_{22}. The constraint is exact: closedness is enforced by orthogonal projection onto $\ker d_1$, and the correction subtracts $\rho v/3$ exactly. No relaxation parameter appears in this volume.
Handed over s_{22}. An acceptance table that bounds local residuals settles the constraint without saying whether it is enforced or relaxed, and a finite expansion recorded at its constant and factor, the bare monomial refused, is a statement of finiteness (constructed for the review).
Written, not decided. $c_{22} \leftrightarrow s_{22}$.

Entry 23. *Obligation.* A smooth curve through data is evidence only when stated as an explicit interpolant with its knots and error; an observation or a knot alone is not a derivation, and only the interpolant's fact is compared with the fact on file.

Citation. Section 5.7: the inherited estimate $\|u - f\|_\infty \le \frac{7}{4}\Omega_f(h) + \frac{5}{2}\epsilon + \frac{h^2}{4\sqrt{6}}\|r\|_2$, with its three terms named, its hypotheses retained, and the statement that coordinatewise accuracy establishes neither a cell attachment nor a sheet nor a period.
Calibration. Standard, cited, and inherited with its hypotheses; not proved here.
Constructed c_{23}. The interpolant is the Hermite spline on a uniform grid of spacing h; its fact is the displayed three-term bound.
Handed over s_{23}. Interpolation named by pages of derivations is a smooth curve through data, and a scalar-trace certificate to be converted is an estimate before the conversion (constructed for the review).
Written, not decided. $c_{23} \leftrightarrow s_{23}$.

Entry 24 (presence). *Obligation.* Rhetorical framing may appear in the exposition but is granted no evidential content: no claim rests on a slogan, and nothing is extracted from it.
Citation. Remark 2.3: the alternating count does not identify a space, no proof uses a figure, and a named object is the object of its definition. Section 1.4: a drawing is the image of a map, defined independently of the drawing.
Calibration. Presence (empty fact extractor). The phrase "looks like" occurs in this volume only as a question, in the epigraph; the propositions it abbreviates are the claims. The source's picture of a hidden hole is named only in Appendix A, as the source's picture, and no claim of this volume rests on it.
Pair. **id** $\leftrightarrow$ **id**: the same symbol on both sides, written and not decided; no fact is extracted.

Entry 25. *Obligation.* Adherence to a canonical method is not itself a result: reduce the appeal to canon to the concrete theorem actually used and compare that with the fact on file; name-dropped theorems contribute nothing.
Citation. Lemma 6.1: nearest-image decoding by the triangle inequality with strict $s/2$, stated as the whole of the decoding theorem used. The proof of Corollary 6.4.
Calibration. Proved here.
Constructed c_{25}. The theorem used is the triangle inequality on a finite image with minimum separation s; no property of the encoding beyond s enters.
Handed over s_{25}. Devotion to canonical texts, and a standard theory cited by its author's name, establish the instance used (constructed for the review). This volume states and proves the instance it uses.
Written, not decided. $c_{25} \leftrightarrow s_{25}$.

Entry 26. *Obligation.* The invariant must be stated as a theory with a checkable remainder, a residual or prediction that could fail, not merely as a hypothesis or a repeated observation; only the theory-level statement contributes a fact.
Citation. Proposition 5.1; Section 5.3, where $\rho = x + y - z$ and $\|e\| = |\rho|/\sqrt{3} = \|d_1\alpha\|/\sqrt{6}$; Section 6.6, where the raw record has $\rho = 3/1000$ and the removed component attains the gap bound exactly.
Calibration. Proved here.
Constructed c_{26}. The remainder is ρ. The theory predicts $\|e\| = |\rho|/\sqrt{3}$ and could fail on a complex with more than one positive eigenvalue; on the torus it is attained.
Handed over s_{26}. A repeated observation, a hypothesis called a law, or the remark that a tolerance relation need not be transitive is a checkable remainder (constructed for the review).
Written, not decided. $c_{26} \leftrightarrow s_{26}$.

Entry 27. *Obligation.* A proposition called true must come with the path by which it became

known; the fact on file is compared with the fact at the end of that path, and an undocumented or deferred path contributes nothing.
Citation. Section 5.6, after Proposition 5.5: similar formulas in different representations do not establish a common observable. Section 3.4: a Boolean sign schedule on a tape becomes an algebraic action only after a representation assigns its symbols. Appendix A: a shared name does not substitute for a map between constructions.
Calibration. Proved here.
Constructed c_{27}. Every truth asserted in this volume carries its path: a proof in the text or a citation. Theorem 7.1 lists the inverse that makes each description true of D.
Handed over s_{27}. A curve fit becomes knowledge through a ledger, and sign schedules are Clifford signatures, a truth cited by name (constructed for the review). This volume calibrates the second identification proposed.
Written, not decided. $c_{27} \leftrightarrow s_{27}$.

Entry 28. *Obligation.* Report the content of the observation rather than the credential of the observer: reduce testimony to the ledger entry it records; testimony that only cites a proof or defers contributes no fact.
Citation. The caption of Table 4.1: equal fiber cardinalities do not identify state spaces or inverses. The closing paragraph of Section 3.3: agreement in the number of retained bits does not identify their contents. The paragraph after the bit in Section 6.3.
Calibration. Proved here.
Constructed c_{28}. The content of an observation is its fiber and the receipt that inverts it, named per observation. Two one-bit receipts, σ and b, have different contents.
Handed over s_{28}. An obvious pattern is a witness, and two receipts counted as two bits are reported with their contents left to the table (constructed for the review).
Written, not decided. $c_{28} \leftrightarrow s_{28}$.

Entry 29. *Obligation.* Separate "the quantity is real" from the announcement that it is: the status counts only when the doubt-resolution chain terminates in a recorded entry; a bare proof-citation or an unresolved doubt contributes nothing.
Citation. Section 7.2: a wrong sheet can produce zero projected discrepancy and still recover a different state. Section 6.4: no decrease of a continuous observation error resolves an exact identification. The closing paragraphs of Section 5.7.
Calibration. Proved here.
Constructed c_{29}. A recovered state is the output of a stated inverse on a stated receipt. Agreement of projections announces nothing about the sheet; the doubt is resolved only by the retained sign.
Handed over s_{29}. A quantity announced real is real, and a frinkahedron announced with the topology of S^3 has that topology (constructed for the review). This volume proves the topology.
Written, not decided. $c_{29} \leftrightarrow s_{29}$.

Entry 30 (presence). *Obligation.* Every universally quantified statement must be accompanied by an admissible local instance, a witness at a point; absent one it is recorded as present, carries no evidential weight, and introduces no new global theorem.
Citation. Section 4.1: a statement about a cycle is a statement on a domain. Section 1.1: a statement quantified over the admitted sector is complete on that domain. The nested-domains paragraph of Section 7.3.
Calibration. Presence (empty fact extractor). The universal statements of this volume are over the finite sector or a declared domain; where an instance is claimed, the worked state is its witness; no

global theorem is added here.
Pair. **id** $\leftrightarrow$ **id**: the same symbol on both sides, written and not decided; no fact is extracted.

Entry 31 (presence). *Obligation.* Logical validity may be recorded as present, the inference rules used, but is granted no empirical content: the chapter marks where deduction ends and observation begins and derives no measured claim from deduction alone.
Citation. Proposition 4.2 and Remark 4.3: the proposition is algebraic, establishes no measured fact, and marks where deduction ends and observation begins. Section 3.4: negative generator squares prescribe no time direction.
Calibration. Presence (empty fact extractor). Deduction is present throughout; no measured claim is derived from it alone.
Pair. **id** $\leftrightarrow$ **id**: the same symbol on both sides, written and not decided; no fact is extracted.

Entry 32. *Obligation.* Replace the assertion "the procedure terminates" by an exhibited finite trajectory: state the intended path through the domain, exhibit the path actually taken, and compare them as data; a non-terminating branch contributes no fact.
Citation. The closing paragraph of Section 6.1: a seed and a retained word determine the state, and the converse fails. Section 6.5: the two-step word from the seed $(1, 0)$ with increments $(-1, +1)$ and $(1, +1)$. Equation (5.2).
Calibration. Proved here.
Constructed c_{32}. The route submitted is the word: S_- with parameter $-\theta$, then S_+. The route taken is the sequence of increments, summing to $(0, 2)$ and reaching $(1, 2)$. Compared as data, they agree.
Handed over s_{32}. Declared halting is an exhibited trajectory, and the instruction that words in generators retain their order is a route (constructed for the review).
Written, not decided. $c_{32} \leftrightarrow s_{32}$.

Entry 33. *Obligation.* Show the measured quantity is invariant under re-presentation at a different level of description: the value on file equals the value re-read from the measurement's own distance record.
Citation. Proposition 2.6 and Remark 2.7: raw radial normalization does not carry the seam to the Clifford torus, and the factorwise map G enforces equal component radii. Section 4.6: under isometric transport the spectrum, the harmonic representatives, and the response γ are unchanged.
Calibration. Proved here.
Constructed c_{33}. The state $(1, 2)$ reads the same from E_F, from E, from the integer code, and from $h_{1,2}$; under Ψ the seam is the Clifford torus; under S_k the constants λ, γ are preserved.
Handed over s_{33}. Raw normalization $p \mapsto p/\|p\|$ is the sphere map (constructed for the review). Retold at that level, the seam need not be carried to the Clifford torus, by the remark cited.
Written, not decided. $c_{33} \leftrightarrow s_{33}$.

Entry 34. *Obligation.* Distinguish "the computation type-checks and terminates" from "the computation establishes the claim": present the computation's output as a checkable proposition and show its content equals the claim, independently of the name attached to the artifact.
Citation. Section 6.7: agreement of representations is a correspondence assertion, not evidence of independent acquisition, and a terminated calculation establishes the proposition it computes, compared on its own terms. Section 7.2, fourth paragraph. Appendix A: recorded source identities are not a report of a build.
Calibration. Proved here for the finite statement; the extension beyond it is proposed.
Constructed c_{34}. The content established is Theorem 7.1 on D. An independent finite calculation,

named in the appendix, reproduces the same propositions; its agreement is checked proposition by proposition, not by the name of its output.
Handed over s_{34}. A compiled file is evidence by its name, and the entire device from which the questions came has this realization (constructed for the review). This volume does not establish the second.
Written, not decided. $c_{34} \leftrightarrow s_{34}$.

Entry 35. *Obligation.* Exhibit the initial datum from which the announced value was computed and show the final number is a function of that datum, not a free-standing announcement.
Citation. Proposition 5.5: one specified q on one admitted domain. Section 1.2: agreement of two readings is a statement about their common observable and encoding assumptions. Theorem 6.3: the exact conclusion uses the operation bound, the observation bound, the sheet, and the admitted alphabet.
Calibration. Proved here.
Constructed c_{35}. The initial data are the seed $(1, 0)$, the word, and the error inputs 10^{-4} per operation and 10^{-3} for quantization. The announced numbers 0.0012, 0.0024, and the bound 0.001374, and the state $(1, 2)$, are functions of those data by the displayed formulas.
Handed over s_{35}. A constant announced from a tape reading, and the thirteen count, are free-standing numbers whose relation to an initial datum need not be exhibited (constructed for the review).
Written, not decided. $c_{35} \leftrightarrow s_{35}$.

Entry 36. *Obligation.* State the submitted claim once, verbatim, as the fixed object of custody; every subsequent answer is addressed to that statement and nothing in it may be amended retroactively.
Citation. Section 1.1: the admitted domain is fixed once and every later statement is addressed to the thirty-six states so admitted. Section 6.1: the rule of admission. Equation (1.2).
Calibration. Proved here; the custody statement is a definition.
Constructed c_{36}. The submitted claim of this volume is (1.2) on the admitted domain, namely that the recovery identity $\mathcal{L}(O(E(s)), R(s)) = s$ holds for every admitted state s, with D the thirty-six states. Every chapter is addressed to it, and nothing in it is amended.
Handed over s_{36}. The object is one whole configuration represented by a frinkahedron with the topology of S^3 (constructed for the review). This volume's custody begins instead with $F = \partial(P_6 \times P_6)$ and proves the topology rather than submitting it.
Written, not decided. $c_{36} \leftrightarrow s_{36}$.

8.3 The lifts

Definition 8.2 (Lift)**.** Page $k + 1$ of the typeset record carries the register of page k lifted to a universe at least as high, and lowering it returns the register unchanged: lowering after lifting is the identity. A *lift* is that passage. It adds no datum to the register; it points to the step taken between the two pages, and the step's mathematics is performed at the pointer. Thirty-five lifts separate the thirty-six pages. What a lift is, a level set once and carried, is fixed in *The Infinite Loop Considered Harmless*, section "The search outside the record".

The press check lowers the register thirty-five times and does none of the mathematics. This section performs lifts 1 to 20, from the fact to the coupling: what each step retains is this volume's question. The numerical card performs lifts 21 to 35, from the coupling to the press. Read in order, the lifts are the verses the typeset record prints, one a page.

Lift 1: the fact to the tape. A fact is a proposition carried with its decision; $T = T$ is decided by reflexivity. A mark is the fact written in a cell. If no field of a mark refers to its carrier, copying the fields is a bijection between marks on any two carriers, its inverse being the copy back, and a mark and its copy agree in every field. The mark fits any carrier; the carrier is retained beside it, not in it. *Proved here*, as in *The Commutator of Representation*, "A mark fits any carrier".

Lift 2: the tape to three readers. For a tape of n marks with click bits $b_0, \ldots, b_{n-1}$, the tally is $T = n$, the register $R = \sum_i b_i 2^i$, and the click count $C = \sum_i b_i$. When every mark clicks, $R = \sum_{i<n} 2^i = 2^n - 1$, $C = n$, and $R + 1 = 2^T$. The fibre of T at n is every tape of length n; the pair (T, R) separates click sequences, by uniqueness of binary expansion of length n; no reader of click bits separates two tapes whose marks click alike and say different things. Each reader is an observation in the sense of Definition 4.1, and what it forgets is its fibre. *Proved here.*

Lift 3: the readers to their agreement. Let ρ overwrite every cell with the truth symbol. Then $T(\rho t) = T(t)$, so every reader $F = g \circ T$ satisfies $F(\rho t) = F(t)$: re-lettering is free for every reader that reads length. Two such readers agree exactly when the lengths agree. *Proved here.*

Lift 4: agreement to cost. Writing one mark sends a tape of length n to one of length $n + 1$, so the counter of Section 8.1, increased by one per entry, is the tally of the writing. Over the field with one element counting costs nothing; writing costs one step a mark. *Proved here*; the cost of a step on a machine is *measured* in the middle card.

Lift 5: cost to the throw. Compare pairs of naturals by $(a, b)\,\mathcal{R}\,(c, d)$ if and only if $ad \leq cb$. For any two pairs, $ad \leq cb$ or $cb \leq ad$, so every two pairs are related and the equivalence generated by $\mathcal{R}$ has one class. Every function out of the quotient is constant: the box has one point. This is the extreme case of Section 4.2: a quotient generated by a total order. *Proved here.*

Lift 6: the throw to the quarter. Since the one-point quotient determines nothing, recovery needs a datum the quotient does not carry: a receipt in the sense of Definition 1.1, with $(O \circ E, R)$ injective. If the datum is carried by every record only as a projection and determined by none, supplying it is not a procedure of the records: it selects. The selected datum is the quarter. *Proved here.*

Lift 7: the quarter to the square. Let O observe and U transform. A map V with $O \circ U = V \circ O$ exists exactly when U sends each fibre of O into a fibre of O: define $V(Ox) := O(Ux)$; it is well defined if and only if $Ox = Ox'$ implies $O(Ux) = O(Ux')$, and it is then unique on the image of O. The clock intertwining (3.4) is such a square, closed. *Proved here.*

Lift 8: the square to the four arms. Compare two truths a, b by what they are, and compare the marks that carry them by how they click. Both 2×2 tables have the click on the diagonal, $T = T$ and $F = F$, and no click off it, $T = F$ and $F = T$; they agree cell for cell and are unchanged when turned over. The four cells are the four arms of the table. *Proved here*, as in *The Commutator of Representation*, "Same difference".

Lift 9: the arms to thirteen. A cell compares two truths by value, and by value $T = F$ and $F = T$ both fail (Lift 8). An arm stores a fact. The arms $T = T$ and $F = F$ store their comparison; the arm false-to-true stores TRUE, read by meaning as the implication from false to true, which

holds; the arm true-to-false stores FALSE. Each arm whose stored comparison can click carries the coefficient tape of the step that sends the local model along the yarn: $3 + 5 + 3 + 2 = 13$ marks. *Measured*, from the inspected declarations of the middle card.

Lift 10: thirteen to five. Every arm carries the frame: a fabric of two crossed strands, a fibre of two strands before them and one after, $2 + 2 + 1 = 5$ marks. The prover's census of the stations of the smooth model and of the yarn names five each time, and the three tapes agree mark for mark. *Measured*, from the inspected declarations.

Lift 11: five to the rule. The arm $T = F$ stores a comparison that cannot click, and the rule takes the thirteen from it. The rule is not derived. It is a bent card in the sense of *The Infinite Loop Considered Harmless*: a datum adjoined from outside, checked where it is used and not computed. *Measured* as to what the middle card declares; the declaration itself is the knower's to accept.

Lift 12: the rule to the table. The five remains on the arm that cannot click; the other three arms weigh $13 + 5 = 18$. The table is

$$M = \begin{pmatrix} 18 & 5 \\ 18 & 18 \end{pmatrix}.$$

Measured, as the weights; the arrangement of the arms is the one of Lift 8.

Lift 13: the table to a count and a clock. Write $M = 18I + N$ with $N = \begin{pmatrix} 0 & 5 \\ 18 & 0 \end{pmatrix}$. The count $18I$ stays on its vertex and reads every T as T. The clock N has no diagonal: each of its steps crosses between the two vertices, one way through the arm $T = F$. *Proved here.*

Lift 14: the clock taken twice.

$$N^2 = \begin{pmatrix} 0 & 5 \\ 18 & 0 \end{pmatrix}^2 = \begin{pmatrix} 90 & 0 \\ 0 & 90 \end{pmatrix} = 90\, I.$$

Two crossings bring every walk home, multiplied by ninety. Since N is off-diagonal, every odd power of N is off-diagonal and $\operatorname{tr} N^{2m+1} = 0$: closed walks of the clock have even length. *Proved here.*

Lift 15: the clock to the root. I and N commute and $N^2 = 90I$, so the eigenvalues of N are $\pm\sqrt{90} = \pm 3\sqrt{10}$ and those of M are $18 \pm 3\sqrt{10}$. The count grows at

$$\lambda_+ = 18 + 3\sqrt{10} = 27.486\,832\,98\ldots$$

and $\lambda_+/13$ is the larger root $(18 + 3\sqrt{10})/13$ of $13y^2 - 36y + 18$, the quadratic of Section 4.2. With the gain $324/5$ of the numerical card, section "The residual, its branch, and its units", the reading is

$$\tfrac{324}{5} \cdot \tfrac{\lambda_+}{13} = \tfrac{972(6+\sqrt{10})}{65} = 137.011\,290\,5\ldots$$

Proved here, given the gain; the gain is *standard* in this volume, cited from the numerical card.

Lift 16: strike the arm, or write it. Give the arm $T = F$ the weight w. Then $N_w = \begin{pmatrix} 0 & w \\ 18 & 0 \end{pmatrix}$, $N_w^2 = 18w\, I$, and the roots are $18 \pm \sqrt{18w}$. Struck, $w = 0$: $N_0^2 = 0$, the larger root is 18, and the reading is $\frac{324}{65} \cdot 18 = 89.723\,076\,9\ldots$, rational, every T read as T. Written, $w = 5$: the reading is $137.011\,290\,5\ldots$, and the whole difference, $\frac{972\sqrt{10}}{65} = 47.288\,213\,6\ldots$, is the clock. *Proved here.*

Lift 17: the clock to a phase. Read the two vertices of N as the two values of a phase, and each crossing as half a turn. By Lift 14 a closed walk crosses an even number of times, so it turns a whole number of times. The parity is *proved here*; that the slot is the phase, half a turn a crossing, is a *proposed interpretation.*

Lift 18: the phase to the spinor. Let $U(\theta) = \exp(-\frac{i\theta}{2}\sigma_3) = \mathrm{diag}(e^{-i\theta/2}, e^{i\theta/2})$. Then $U(2\pi) = -I$ and $U(4\pi) = I$: once around flips, twice around returns. In the quaternion picture of Section 3.4, q and $-q$ induce one rotation, so the lost sign is invisible to every observation that factors through the rotation: $O(-\psi) = O(\psi)$. The sign is a relative phase, retained only by a receipt, and not a failure of the square to close. *Proved here*; the double cover is *standard.*

Lift 19: the spinor to an observable.

$$U(\theta)\sigma_1 U(\theta)^* = \cos\theta\,\sigma_1 + \sin\theta\,\sigma_2, \qquad U(\theta)\sigma_3 U(\theta)^* = \sigma_3.$$

The off-diagonal observable σ_1 rotates and returns exactly when $\theta \in 2\pi\mathbb{Z}$, at the quarter-turns $\theta_k = k\pi/2$ exactly when four divides k; σ_3 holds still at every angle, the control. *Proved here*, by multiplying the diagonal matrices.

Lift 20: the reading to the coupling. Take the reading of Lift 15 for the coupling of light to the electron,

$$\frac{e^2}{4\pi} = \frac{1}{137.011\,290\,5\ldots}.$$

This is a bent card: no step before it derives it. *Proposed interpretation.* From here the numerical card carries the coupling through transport to the stationary system of light and matter, and to the laboratory.

8.4 The press check

Definition 8.3 (Press check)**.** The *press check* is the decidable function

$$\mathrm{PressCheck}(\mathrm{Reading}_A, \mathrm{Reading}_B) = (\mathrm{Slipped},\ \mathrm{Value},\ \mathrm{Velocity}), \qquad \mathrm{Slipped} = \neg(\varphi_A \leftrightarrow \varphi_B),$$

where φ_A, φ_B are the facts that two independent retrieval pipelines recover the register, Value is the register read back, and Velocity is the top value of the counter. It is evaluated once, at the head of the record.

The slip compares each reading with the certified register, not the two readings with each other: a pair slips exactly when one reading recovers the register and the other returns a different state. A pipeline that returns no state supplies no reading, and then there is no pair to check; that outcome is not a slip. The press check reads only the head; the separate reading of the calibration column follows the verdict.

Reading A: orthogonal observation with the sheet sign. By (6.2), the worked state $(1, 2)$ has observation

$$x = \left(\frac{1}{2\sqrt{2}}, \frac{\sqrt{3}}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}} \right), \qquad w = \frac{\sqrt{3}}{2\sqrt{2}} > 0,$$

so the retained sheet sign is $+$. The lift $L_+(x)$ of (2.2) returns (x, w) exactly, and the inverse of E in Section 6.1 reads the exponents 1 and 2 from the two complex coordinates. Reading A recovers $(1, 2)$. Under the declared bounded errors the same recovery holds: the lifted error is at most $3/1250$, below the decoding radius $1/(2\sqrt{2})$, by Section 6.5 and Corollary 6.4.

Reading B: paired Hopf observation with the antipodal bit. By Section 6.3, the worked state has paired phases $(u, v) = (3, 5)$. Equation (6.3) gives $j_0 = \frac{1}{2}((3 + 5) \bmod 6) = 1$ and $k_0 = (3 - 1) \bmod 6 = 2$, and the retained bit $b = 0$ selects (j_0, k_0). Reading B recovers $(1, 2)$. The paired image has separation $s_H = \sqrt{2}$, Section 6.4, so the exact decoding tolerates numerical error below $1/\sqrt{2}$.

The verdict on the pair. Here φ_A is "Reading A recovers the register" and φ_B is "Reading B recovers the register". Each is decided true above by displayed arithmetic, and neither decision uses a choice. The equalizer $\neg(\varphi_A \leftrightarrow \varphi_B)$ is the exclusive disjunction, false since both readings recover the register:

$$\textbf{Verdict} = (\textbf{false},\ ((1, 2), 13, +, 0),\ 36).$$

The demand-level reading. The entries write $c_k \leftrightarrow s_k$ without deciding it, and the press check reads nothing below the head. A second reading, outside the press check's rule, reads the calibration column of the trace: not the written propositions, which stay undecided, but the calibration each entry assigns to its citation. It records whether s_k is a proposition this volume has established. At three entries s_k exceeds the proved domain D and is audited *not established here*:

1. entry 34, the proposed extension of the finite correspondence to the entire device from which the questions came, whereas Theorem 7.1 concerns D and its augmented descriptions and the appendix states the larger domains explicitly;
2. entry 27, the proposed identification of a Boolean sign schedule on a tape with a Clifford signature, whereas Section 3.4 obtains an action only after a representation assigns the symbols;
3. entry 10, every physical reading of the configuration space, the de Sitter reading of Section 7.4 included, whereas the coordinates remain configuration coordinates until calibrated, by the opening of Chapter 2 and the close of Section 7.3.

At every other entry the citation is a theorem, a definition, or a cited result of this volume, and the record claims only what it cites. This reading decides no $c_k \leftrightarrow s_k$; the verdict *slipped* is the press check's alone.

Inhabitation is not acceptance. The existence of the record, thirty-six entries each with a citation, carrying one register through a counter that reaches thirty-six, is well-formedness and is independent of the verdict: the same form carries a state read with the wrong sheet and then reports slipped. Inhabitation is well-formedness of the trace; acceptance is the truth value of the equalizer evaluated at the head.

8.5 The hypothesis the apparatus tests

The review and the typeset record are the apparatus of a hypothesis stated earlier in the middle construction. Without its apparatus, and calibrated proposed, it is the following. Names are required by a proof and are never produced by one; a construction advances one name per rung. A representation is optimally partitioned only when its higher-order terms vanish. And there is a representational noise floor: below it, more symbols would be required to tell a falsifiable repeat of an observation from an unfalsifiable exact replay of its record, so that a hypothesis which can be falsified needs noise, lest its repeats be exact, observable cycles that return the record to itself.

This volume does not test the hypothesis. It identifies its own closest realization of the floor. With N positions per clock the least state separation is $s_N = \sqrt{2}\sin(\pi/N)$, Section 7.3. Exact recovery of a discrete label tolerates a state error strictly below $s_N/2$, and that allowance shrinks to zero as N grows, while the continuous inverse of Proposition 2.10 remains valid on a fixed sheet margin. The clock words are periodic: each generator has order six, and the generated group has eighteen elements, so an exact replay of a word returns the state exactly and leaves a record identical to the first. Two physical repetitions can also leave identical records, so equality of records does not tell a repeat from a replay. A repeat whose record differs from the first carries the difference in its errors, and it remains decodable only when those errors lie below the allowance. The errors that are both visible in the record and decodable form the open interval from zero to $\sin(\pi/N)/\sqrt{2}$, and that window closes as the alphabet is refined. The declared inputs of Section 6.5 lie in it at $N = 6$; at a sufficiently refined alphabet the same inputs would not. The separation and decoding bounds fix the window. They do not show that a testable repeat requires nonzero error; that is a further clause of the hypothesis, and the identification of the window with the representational floor is proposed, as the ledger records.

Appendix A

Sources and the episode correspondence

A.1 Origin of the construction

The term *frinkahedron* and the questions about hidden enclosure, counting, and retained readings originate in the *Measurement* episode manuscripts. The root document `SPECIFICATION.md` sets out the geometric program. The earlier instrument study develops the observation, transport, local correction, period, and information framework. The present monograph selects an explicit product polytope and finite state sector to make the geometric realization complete on a stated domain.

The construction contributes the vertices, prism facets, incidence relations, seam-preserving sphere map, circle actions, and thirty-six-state encoding used in the text. These are definitions followed by mathematical proofs. Their relation to the local source passages is recorded below so that a shared name does not substitute for a map between constructions.

A.2 Episode concordance

The sources were inspected on 22 October 2026. Paths in this section are relative to `CARD_2/Measurement/`. Declarations are cited by name. Line numbers are not recorded: they change with every edit of a source file, whereas the declaration names are the stable anchors.

A.2.1 The enclosing picture

`Episode03.lean`, in the commentary beside its ordering relation on paths, discusses the hidden-hole picture and draws the frinkahedron; the passage is prose, not a declaration. `Episode06.lean`, in its opening commentary, lists twelve conceptual fenceposts. The twelve prism facets of Chapter 2 are the facets of the selected product polytope. The text proves their incidence structure directly. The episode passages motivate the enclosure question but do not themselves provide this inventory of polygon attachments.

A.2.2 Once and twice around

`Episode05.lean` declares the `once_around` and `twice_around` fields of type `SpaceTimePath`. `Episode08.lean` defines `once_around_the_integral` and `twice_around_the_integral` and proves `once_around_holds`, `twice_around_slips`, and `the_two_slips_meet`, which concern integer counts and signs of a quantity it calls `the_slack`. The two circle actions in Chapter 3 are

defined by their phase maps. An identification with a source path datatype would require a state map and an intertwining proof for that datatype. The finite clock sector supplies such a proof for its own generators.

A.2.3 The retained tape reader

`Episode08.lean` defines `the_register` by a Horner fold over a list of facts and states its all-ones and monomial identities. This is an exact reader of a retained tape. The integer code $j + 6k$ of Chapter 6 is another exact reader with its own inverse. It reads two six-position indices. A comparison with the tape reader would specify a tape encoding of those indices and prove agreement after applying the selected readers. The count thirteen that the same file reaches through `through_thirteen` is a separately defined quantity; the worked state's code $13 = 1 + 6 \cdot 2$ in Section 6.5 is not identified with it.

A.2.4 A numerical receipt beside a quotient

`Episode05.lean` defines `TruthOrder` and `FACT.SAME`. `Episode08.lean` defines the cross-multiplied `ReadingOrder`, proves `any_pair_is_its_inferred_integer`, and defines `theReadingCertified`, which retains an integer together with a quotient equality. Proposition 4.2 explains the general mathematical distinction: a descended reader is constant on each quotient class, while an augmented output may retain a representative that the quotient alone has forgotten. The finite constant-quotient example exhibits this distinction explicitly.

A.3 The review and the typeset record

`Episode10.lean` declares the class `REVIEWED`, indexed by a page number and holding one field, `rebuttal`. It gives thirty-six instances, one per page from 36 down to 1, each named with the prefix `REVIEWER_2_`. The instance for page 36 admits the manuscript unchanged. Each instance for a page below 36 takes the instance for the page above it and prepends one methodology cell quoting the fact found by search (the constructed side) and the fact handed over by name (the slip side). Three pages, 31, 30, and 24, write the carrier's symbol on both sides and extract nothing. Chapter 8 answers page n in reply entry n of Section 8.2; Table A.1 records the map. The file order of the instances is not monotone in the page number; the chain is linked by each instance's dependence on the page above it, and every page from 1 to 36 occurs exactly once.

Table A.1: Review pages, their instance names, and the reply entries that answer them.

Page	Instance (suffix after `REVIEWER_2_`)	Reply
36	`ASKS_WHERE_THE_RECORD_BEGINS`	entry 36
35	`ASKS_WHAT_THE_ANSWER_WAS_ANSWERING`	entry 35
34	`ASKS_IF_COMPILING_IS_EVIDENCE_OR_A_FILENAME`	entry 34
33	`ASKS_HOW_THE_NEEDLE_SURVIVED_THE_RETELLING`	entry 33
32	`ASKS_IF_HALTING_WAS_CHECKED_OR_DECLARED`	entry 32
31	`CATCHES_LOGIC_POSING_AS_AUTHORITY`	entry 31, presence
30	`REFUSES_A_UNIVERSAL_CLAIM_WITHOUT_A_LOCAL_WITNESS`	entry 30, presence
29	`DISTINGUISHES_REALITY_FROM_ITS_PRESS_RELEASE`	entry 29
28	`ASKS_WHAT_WAS_ACTUALLY_WITNESSED`	entry 28

Page	Instance (suffix after REVIEWER_2_)	Reply
27	OBJECTS_TO_TRUTH_CITED_BY_NAME_ONLY	entry 27
26	REQUIRES_THE_INVARIANT_TO_CARRY_A_REMAINDER	entry 26
25	ASKS_IF_DEVOTION_TO_METHOD_IS_A_METHOD	entry 25
24	SEPARATES_PERSUASION_FROM_EVIDENCE	entry 24, presence
23	SUSPECTS_THE_SMOOTH_STORY_IS_INTERPOLATION_IN_A_LAB_COAT	entry 23
22	ASKS_IF_THE_CONSTRAINT_WAS_ENFORCED_OR_PENALIZED	entry 22
21	NOTES_A_LOAD_MUST_LAND_IN_A_BASIS	entry 21
20	REQUIRES_SCALE_TO_SHOW_ITS_MULTIPLICATION	entry 20
19	ASKS_WHERE_MAGNITUDE_BECOMES_ADDITIVE	entry 19
18	DISTINGUISHES_A_VALUE_FROM_ITS_PROCESS	entry 18
17	ASKS_IF_EXECUTION_LEFT_AN_OPCODE_RECORD	entry 17
16	ASKS_WHICH_SOURCE_FACT_SURVIVED_THE_CHANNEL	entry 16
15	MAKES_THE_JOKE_PAY_ITS_EVIDENTIARY_BILL	entry 15
14	ASKS_IF_THE_CLOCK_IS_OBSERVABLE_OR_ASSERTED	entry 14
13	ASKS_WHERE_PRESENCE_ACCUMULATES	entry 13
12	ASKS_IF_OBSERVATION_HAS_A_RECORDED_DIMENSION	entry 12
11	ASKS_WHAT_MAKES_COMPARISON_MORE_THAN_PARALLEL_NAMING	entry 11
10	REQUIRES_THE_PHYSICAL_CLAIM_TO_SHOW_ITS_BASE_FACT	entry 10
9	ASKS_IF_REPRESENTATION_HAS_A_CARRIER_OR_A_NOTATION	entry 9
8	ASKS_WHERE_THE_NUMBER_ACTUALLY_COUNTS	entry 8
7	ASKS_IF_REPETITION_REPEATS_THE_EXPERIMENT_OR_THE_SENTENCE	entry 7
6	ASKS_WHERE_THE_MISSING_EXHIBIT_WENT	entry 6
5	ASKS_IF_RESIDUE_IS_A_RECORD_OR_A_LEFTOVER_STORY	entry 5
4	ASKS_IF_ENCODING_KEEPS_A_READ_PATH_TO_THE_MARK	entry 4
3	ASKS_IF_COUNTING_COUNTS_A_CARRIER_OR_NAMES_A_CLASS	entry 3
2	ASKS_WHO_ADMITTED_THE_MARK	entry 2
1	ASKS_IF_IT_ALL_RESTS_ON_A_MARK_OR_ON_CONFIDENCE	entry 1

`Episode11.lean` declares the class `TYPESET`, indexed by a rung number, with three fields: `register_value`, `universe_id`, and `the_page`. Its instances are `TYPESET_Episode1_FACT` at rung 1 and `TYPESET_page_2` through `TYPESET_page_36`; each instance above rung 1 lifts the register below it once, applies one successor to the counter below it, and prepends one results cell, quoting the head of the rung's review record and the proposition that its two facts are equivalent, to the page below it. The structure `PressVerdict` has fields `slipped`, `value`, and `velocity`, and `pressCheck` computes it from the head of the top page: `slipped` is the negation of the equivalence of the two quoted facts, decided from their own decision procedures. Chapter 8 realizes these fields as follows.

- `register_value` is the register of Section 8.1: the worked state $(1, 2)$ with its receipts.
- `universe_id` is the counter: the reply index. The source counts from zero at its floor, so its top reads thirty-five successors; the record counts replies from one and reports thirty-six. The two conventions differ by the floor's zero and by nothing else.
- `the_page` is the trace: the thirty-six entries of Section 8.2, each quoting its pair and the unresolved equivalence.
- `pressCheck` is Section 8.4: the two readings, the exclusive disjunction, and the triple.

`Episode12.lean` builds the floor of the typeset ladder by hand: `theManuscript` is a three-cell tape decoded from three bits; `reviewedByHand` is a rung-1 review record with that manuscript as its rebuttal; `typesetByHand` is a rung-1 typeset record whose register and counter are both

`cookPage`, the number zero stamped on a fact; and `typesetBase` registers it as the rung-1 instance. Its instance `reviewedAny` is parametric in the page number and returns the same manuscript at every page. It is a gimmicked card: it answers every reviewer page with the identical tape and routes around the thirty-six-page descent, and the file names it as such. With that card in place instance search resolves `theTop`, the rung-36 record, whose page is `theTopTrace`; the theorem `cook_levin_on_the_mechanism` states that this trace satisfies its own step formula exactly when every cell's pair differs. The file does not invoke `pressCheck` on `theTop` as a named theorem; the verdict on that card, slipped true with value zero and velocity thirty-five, is a deduction by unfolding recorded in its commentary. The present volume's record uses no parametric card: each of its thirty-six entries cites a different passage, and its verdict is computed in Section 8.4 from two readings proved in Chapter 6.

A.4 The earlier observation framework

The earlier instrument study is used through specific constructions. `AXIOMS.md` formulates exact observation equivalence and retained recovery. The study's distance chapter develops the scalar trace estimate and the two-projection commutator reproduced in Chapter 5; the trace estimate is the passage headed "complete error budget" in that chapter, which Section 5.7 cites as the complete error budget of the earlier instrument study. Its transport chapter carries metrics and readers together. The measured loop framework in `FOURTH_ARGUMENT.md` and the study's loop chapter motivates harmonic recovery; the symmetry and entropy chapters develop compatible operations and the information lost by compression.

The present text provides its own proofs for the selected sphere, finite fiber counts, torus matrices, inverse estimates, and worked finite recovery. The scalar trace estimate is identified as an inherited result with its sampling and calibration assumptions retained. External references support the standard topology, Clifford, Hodge, approximation, and numerical-analysis frameworks. They do not replace the model-specific maps proved here.

A.5 Mathematical construction and source evidence

The exact established correspondence is among the finite clock states, their polytope and round encodings, their retained observations, and their period records. Theorem 7.1 states that result. The entire episode datatype and a physical spacetime interpretation have larger domains than this finite construction. Extending the correspondence to either would require the relevant encoding, operations, observation maps, and calibration.

The repository records inspected source identities in a source snapshot. Its hashes identify working-tree bytes; they are not a report of a Lean build. The independent finite calculation `check_model.py` reproduces the stated finite observations and matrix identities. The worked error allowances in Chapter 6 are synthetic deterministic inputs, clearly separated from an apparatus measurement. This provenance keeps the proof of the selected model and the evidence for any later application available for independent examination.

A.6 Stanzas and results

The review's thirty-six pages and the typeset record's thirty-six rungs name the same ladder of classes in the same order, so one concordance serves both. Each row gives the rung, the act its typeset verse names, the demand its review page makes, the kind of item that answers it in this

volume, and the located items themselves. “Named result” means a theorem, proposition, corollary, definition, or table; “worked calculation” means a displayed equation inside a section; “presence” marks the three pages that extract no fact. The full replies, with their constructed and handed-over readings, are in Chapter 8.

Rung	Verse (act)	Page (demand)	Kind	Located items
1	the origin: the carrier's number, zero, the first page	asks if it all rests on a mark or on confidence	section	Section 6.4; Section 6.2; Section 5.7
2	admits the mark	asks who admitted the mark	worked calculation	eq. (6.1); Section 6.1; Section 1.1; Chapter 6
3	counts the number	asks if counting counts a carrier or names a class	named result	Proposition 3.4; Proposition 2.2; Section 6.1
4	encodes the glyph	asks if encoding keeps a read path to the mark	named result	Theorem 7.1; eq. (1.2); Section 6.1
5	residues the code	asks if residue is a record or a leftover story	named result	Proposition 5.1; Section 5.3; Section 6.6
6	binaries the value	asks where the missing exhibit went	named result	Proposition 3.3; Table 6.1; eq. (6.3); eq. (2.2)
7	repeats the process	asks if repetition repeats the experiment or the sentence	section	Section 7.3; Section 4.3; Section 5.7
8	numbers the value	asks where the number actually counts	named result	Proposition 2.2; Table 2.1; Table 6.1; Section 2.1
9	represents the code	asks if representation has a carrier or a notation	section	Section 3.4; Section 6.1; Section 4.6
10	physicals the method	requires the physical claim to show its base fact	section	Section 6.4; Section 7.3; Chapter 2; Section 7.4
11	compares the process	asks what makes comparison more than parallel naming	named result	Proposition 5.5; eq. (5.3); Section 6.5; Section 1.4
12	observes the mark	asks if observation has a recorded dimension	named result	Proposition 2.10; eq. (2.2); eq. (6.2); Section 2.4
13	presents the process	asks where presence accumulates	section	Section 6.6; Section 4.4; Section 2.5.1
14	measures the step	asks if the clock is observable or asserted	named result	Figure 3.2; eq. (3.4); Section 6.1
15	gungans the glyph	makes the joke pay its evidentiary bill	named result	Definition 2.1; Definition 4.1; Table 6.1; Section 1.2
16	sources the code	asks which source fact survived the channel	named result	Proposition 4.4; Section 4.3
17	executes the code	asks if execution left an opcode record	section	Section 3.5; Section 5.5; Section 6.1
18	values the function	distinguishes a value from its process	section	Section 3.5; Section 6.1
19	magnitudes the value	asks where magnitude becomes additive	worked calculation	eq. (5.2); eq. (6.4); Section 5.5; Section 6.7
20	scales the value	requires scale to show its multiplication	section	Section 2.5; Section 2.5.1; Section 5.2

Rung	Verse (act)	Page (demand)	Kind	Located items
21	loads the process	notes a load must land in a basis	named result	Theorem 5.3; eq. (4.2); Section 4.5; Section 5.2
22	finite-elements the model	asks if the constraint was enforced or penalized	section	Section 4.4; Section 4.5; Section 5.4
23	bullshits the model	suspects the smooth story is interpolation in a lab coat	section	Section 5.7
24	propagandizes the model	separates persuasion from evidence	presence	Table 2.1; Section 2.1; Section 1.4; Appendix A
25	acolytes the method	asks if devotion to method is a method	named result	Corollary 6.4; Section 6.4
26	sciences the method	requires the invariant to carry a remainder	named result	Proposition 5.1; Section 5.3; Section 6.6
27	truths the code	objects to truth cited by name only	named result	Proposition 5.5; Theorem 7.1; Section 5.6; Section 3.4
28	witnesses the mark	asks what was actually witnessed	named result	Table 4.1; Section 3.3; Section 6.3
29	reals the model	distinguishes reality from its press release	section	Section 7.2; Section 6.4; Section 5.7
30	locals the variable	refuses a universal claim without a local witness	presence	Section 4.1; Section 1.1; Section 7.3
31	universals the function	catches logic posing as authority	presence	Proposition 4.2; Section 4.2; Section 3.4
32	logics the code	asks if halting was checked or declared	worked calculation	eq. (5.2); Section 6.1; Section 6.5
33	halts the method	asks how the needle survived the retelling	named result	Proposition 2.6; Section 2.2; Section 4.6
34	measures the step	asks if compiling is evidence or a filename	named result	Theorem 7.1; Section 6.7; Section 7.2; Appendix A
35	compiles the tape	asks what the answer was answering	named result	Proposition 5.5; Section 1.2; Section 6.5
36	outputs the value	asks where the record begins	named result	Corollary 6.4; Theorem 7.1; Proposition 2.10; eq. (1.2)

Bibliography

[1] Douglas N. Arnold, Richard S. Falk, and Ragnar Winther. Finite element exterior calculus: from Hodge theory to numerical stability. *Bulletin of the American Mathematical Society*, 47(2):281–354, 2010. Author version: arXiv:0906.4325.

[2] Carl de Boor. *A Practical Guide to Splines*. Springer, New York, 1978.

[3] Carl de Boor and Robert E. Lynch. On splines and their minimum properties. *Journal of Mathematics and Mechanics*, 15(6):953–969, 1966.

[4] Jean Gallier. Clifford algebras, clifford groups, and a generalization of the quaternions, 2008. arXiv:0805.0311; revised 2014.

[5] G. W. Gibbons and S. W. Hawking. Cosmological event horizons, thermodynamics, and particle creation. *Physical Review D*, 15(10):2738–2751, 1977.

[6] Gene H. Golub and Charles F. Van Loan. *Matrix Computations*. Johns Hopkins University Press, 4 edition, 2013.

[7] Allen Hatcher. *Algebraic Topology*. Cambridge University Press, 2002.

[8] Marc Henneaux and Claudio Teitelboim. The cosmological constant and general covariance. *Physics Letters B*, 222(2):195–199, 1989.

[9] Xiaoye Jiang, Lek-Heng Lim, Yuan Yao, and Yinyu Ye. Statistical ranking and combinatorial Hodge theory, 2008. arXiv:0811.1067; revised 2009.

[10] Joint Committee for Guides in Metrology. Evaluation of measurement data: Guide to the expression of uncertainty in measurement. Technical Report JCGM 100:2008, Bureau International des Poids et Mesures, 2008.

[11] Joint Committee for Guides in Metrology. Guide to the expression of uncertainty in measurement: Part 6: Developing and using measurement models. Technical Report JCGM GUM-6:2020, Bureau International des Poids et Mesures, 2020.

[12] David W. Lyons. An elementary introduction to the Hopf fibration. *Mathematics Magazine*, 76(2):87–98, 2003.

[13] Saul Schleimer and Henry Segerman. Sculptures in S^3. In *Proceedings of Bridges 2012: Mathematics, Music, Art, Architecture, Culture*, pages 103–110, 2012.

[14] Erwin Schrödinger. *Expanding Universes*. Cambridge University Press, Cambridge, 1956.

[15] W. G. Unruh. Unimodular theory of canonical quantum gravity. *Physical Review D*, 40(4):1048–1052, 1989.

[16] Steven Weinberg. The cosmological constant problem. *Reviews of Modern Physics*, 61(1):1–23, 1989.

[17] Peter Woit. Clifford algebras and spin groups. Columbia University, Math G4344 lecture notes, 2012. Spring 2012.